



Thinking in a Low-Resource Language: What SFT Builds, What RL Fixes, What Accuracy Cannot See

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August 2026

Abstract

Take three frontier mixture-of-experts models (Alibaba, OpenAI, NVIDIA; 3.6–4.0B active parameters each) and fine-tune them to reason in a low-resource language. On accuracy benchmarks, almost nothing happens, and the benchmark itself is noise at this scale: changing only the random seed moves the score by 7.7 points, more than every data and recipe effect we measured. That null is our first result. The real changes live where accuracy cannot see. Base models never think in Greek: 0 of 1,000 reasoning traces, even when the question is Greek, so the model answers correctly while reasoning in a form its user cannot read, audit, or correct. After supervised fine-tuning (SFT), every released checkpoint reasons in the language of the question on ~98% of items, one family at 3× fewer tokens, with judged grammaticality improving on all four models and general ability within a few points of each model’s own base in both languages: nothing was forgotten, and fluency was gained. We propose the six behavioural dimensions that make such changes measurable (which language the model reasons in, what it spends, whether it can tell easy from hard, and what it forgot), each gated to reject any metric that correlates with output length, and we report how our own instruments lied: six failures, each caught by a control, each of which had already convinced us of something false. What SFT cannot do is fix its own defects: a quarter of answers skip the requested format, answers leak into the reasoning channel, and an explicit “think in English” is obeyed under half the time. Reinforcement learning with verifiable rewards, pre-registered before training, fixes the first two outright (answer-format fallback 24% → 2.5%, answer-channel leak 3.5% → 0.0%, both against a flat random-reward control) and moves the third (+9.1 pp, real but short of its pre-registered bar), while the Greek reasoning habit survives an accuracy-only gradient untouched (98.2% fidelity). We release five checkpoints. The instruments, the controls and the pre-registration travel to any low-resource language; Greek is the case that let us measure them.

1. Introduction: one number, six dimensions

A reasoning fine-tune is judged the way every fine-tune is judged: one accuracy number, before and after. For a low-resource language that number answers a question nobody asked. It does not say which language the model reasons *in*, how many tokens it burns getting there, whether it can tell an easy question from a hard one, or what it gave up in exchange.

The study is designed around the constraint a low-resource deployment actually faces: the serving bill. We hold the *active* parameter count fixed (3.6–4.0B per token) and vary the lab: three sparse mixture-of-experts families from Alibaba, OpenAI, and NVIDIA, each a frontier lab’s different architectural bet on the same inference budget (§2). Sparse MoE is the architecture of interest precisely because it is cheap: a 20–36B-parameter model that serves at the cost of a 4B dense one is what makes local reasoning models economically viable for a

language community that cannot fund frontier-scale inference. Running the same recipe, corpus, and instruments across all three families is what lets us separate what the fine-tune does from what one architecture happens to do.

Greek already has a dedicated open-model ecosystem we build on rather than from scratch: the Meltemi and Krikri instruction models [26, 35], their translated evaluation suites, and our own Sophea-Titan-1 general-purpose model. What none of that line measures is long-form *reasoning* behaviour: those models answer in Greek, but no released checkpoint thinks out loud in it, and no Greek benchmark can currently tell the difference between a model that does and one that plans in English and translates at the end.

We measured all of those, and the contrast is the paper. On accuracy, almost nothing happened (the best of the 15 arms of §7 scores 76.5 against the base’s 77.2, both on the 1,000-item three-axis Greek probe in think mode),

and for several weeks we treated that as a corpus problem and built five more corpora to fix it.

None of them fixed it, and eventually we ran the experiment that should have come first: we retrained one configuration, changing nothing but the random seed. The score moved 7.7 points. Every difference we had been interpreting (better data selection, a better corpus, a changed training schedule) was smaller than that.

This paper is what remains after that discovery, and what replaces it, told in the order the reasoning has to run. Section 2 fixes the models, the corpus, and the evaluation lanes; Section 3 defines the six dimensions before any of them is used. Section 4 establishes the noise floor. Section 5 shows that the properties which *did* change are not accuracy and are not seed-sensitive: the language the model reasons in, the tokens it spends, and whether it can tell an easy question from a hard one. Section 6 asks what the fine-tune forgot. One training-recipe comparison mattered enough to keep as a self-contained side experiment: it explains our answer-format failure mode and is the one accuracy effect that survives, replicated across 15 independently trained arms rather than read off a single run; it is reported on its own terms in §7. Section 8 reports six instrument failures: five controls each of which killed or corrected a finding we believed, and Section 9 states the acceptance rule they taught us. Section 10 then repairs the one defect the recipe installs (the language lock) and Section 11 closes the two remaining pre-registered questions. What follows reads the same evidence per domain (§12) and per family (§13), and Sections 14–16 state what we can and cannot recommend on it.

The claim. We find that for a low-resource-language reasoning fine-tune, accuracy on a translated benchmark is close to uninformative: it is dominated by training noise, distorted by answer-format effects, and inflated by contamination that standard checks miss. The behavioural dimensions are stable, large, and measure what the fine-tune was actually for.

2. Setup

Models. Three sparse mixture-of-experts model families [12, 30] from three different labs (four checkpoints in all, since the Nemotron line contributes two generations), chosen because they occupy the *same serving budget* while making different architectural bets with it:

Total size differs by $1.7\times$ while active size differs by 11%. All three cost roughly the same to serve and spend that budget differently: Qwen routes 8 of 256 experts (3.1% of its parameters per token), Gpt-OSS 4 of 32 (12.5%), NemotronH 6 of 128 while replacing most of its attention with state-space layers. Comparing them on *total* parameters would say Qwen is $1.7\times$ the model that

Gpt-OSS is; comparing on *active* says they are the same size. We report both and treat active as the meaningful axis, because it is what a deployment pays.

This also bounds what our results generalise to: MoE models are not evidence about dense ones (Limitations). And MoE changes what LoRA attaches to.

Adaptation. LoRA [16] at $r=32$, $\alpha=64$, one epoch, effective batch 32, learning rate 2×10^{-4} . All training, merging, and evaluation ran on a single NVIDIA DGX B200 node: $8 \times$ B200 GPUs (180 GB HBM3e each, 1.44 TB aggregate), which fits every configuration in this paper without model parallelism beyond FSDP sharding, a deliberate constraint, since a recipe for a low-resource language should be reproducible on one node. *Expert stride 3* means adapters are placed on every third MoE layer rather than all of them; the shared expert, which lies on every token’s path, is always included. The stride exists because expert count, not model size, sets the number of adapters: where a model stores its experts as stacked parameters, PEFT attaches one adapter per stack (28 for Qwen, 16 for Gpt-OSS), but where they are individually-materialised `nn.Linear` modules it attaches one per expert: 4,188 separate adapter tensors on a model we profiled, which costs step time in kernel launches and gradient all-reduces rather than in FLOPs. Fused-expert LoRA also cannot be loaded back by `PeftModel.from_pretrained` under our library versions, so every Qwen checkpoint is merged into dense weights before evaluation.

Corpus. 118,092 Greek rows in two halves of almost equal size.

The *reasoning half* (59,107) carries an explicit trace in a separate field. It is 98.5% **synthetic**: questions and gold answers come from public English datasets, but the traces are generated (see below). Only 894 rows carry a trace we did not produce (the domain mix is Table 2).

The *direct half* (58,985) has no traces. It is the instruction corpus of **Sophea-Titan-1**, a previously released Greek model, reused unchanged: Aya [33] (33.8%), a Greek instruction set (25.8%), synthetic multi-turn dialogue (10.1%), Greek QA (7.7%) and twelve smaller sources, plus $\sim 2,600$ rows of domain chain-of-thought (legal, medical, finance, energy) and 50 identity rows. It is 91% Greek and 9% English (5,211 rows), the English kept deliberately as replay against catastrophic forgetting. Reusing a corpus that had already produced a working Greek model is why we did not initially suspect this half of anything.

Its intended role is to keep the model’s non-reasoning mode alive, and it does: §7 shows that omitting it collapses the reasoning switch. It is also, on the same evidence, what costs the model most of its accuracy and its answer-format compliance. That tension is the paper’s one replicated recipe finding.

lab	base model	total	active	routing	fine-tuned release
Alibaba	Qwen3.6-35B-A3B	36.0B	3.97B	8 of 256	Sophea-Qwen3.6-v1
OpenAI	Gpt-OSS-20B	20.9B	3.60B	4 of 32	Sophea-OSS-v1
NVIDIA	NemotronH-30B-A3B	31.6B	3.58B	6 of 128	Sophea-Nemo-3-Nano-v1
NVIDIA	Nemotron-3.5-Lightning-30B-A3B	31.6B	3.58B	6 of 128	Sophea-Nemo-3.5-Lightning-v1

Table 1: Total and routed-expert parameters counted from the released tensors; active-per-token is non-routed + k/E of routed. Gpt-OSS ships MXFP4-packed experts, so its figures are vendor-reported. NemotronH (Nano) and Nemotron-3.5-Lightning are **Mamba/MoE hybrids** (Mamba, MoE and attention layers of 52; the state-space line of Dao and Gu [8], Gu and Dao [14] and the vendor’s Nemotron-H report [21]); the other two are MoE transformers. The right-most column names the fine-tuned Sophea reasoning release each base produces; all four SFT releases are language-matched, and a fifth release, Sophea-Qwen3.6-v1.1, is the RLVR refinement of the Qwen row (§15.1). Throughout, *base* refers to the vendor checkpoint and *fine-tuned* to its Sophea reasoning model. Lightning is the next-generation release in the Nemotron line; Nano and Lightning share the 31.6B / 6-of-128 routing profile and are reported together as the Nemotron family (Nano is the same-generation member of that family).

domain (reasoning half)	rows	share
mathematics	22,608	38.2%
science	11,064	18.7%
world knowledge	5,884	10.0%
deductive logic	5,151	8.7%
medical	4,898	8.3%
commonsense	3,904	6.6%
reading comprehension	3,828	6.5%
physics	1,770	3.0%

Table 2: The reasoning half is math-heavy and thin on the axes we evaluate hardest: commonsense is 6.6% and logic 8.7% of it.

Where the traces come from. Questions and gold answers are taken from public English datasets and translated; the *traces are generated*, not translated. Asking a translator to render a chain of thought returns a tidy summary (which is the defect described next), so we prompt an LLM to solve each question afresh in Greek and keep the trace only if its final answer agrees with the gold. The generators are two frontier commercial models from a single family: one for the mathematics slice (from OpenR1-Math-220k questions) and a larger one for the logic, commonsense and regenerated-ECQA slices); the remaining reasoning rows are carried unmodified from their sources (Llama-Nemotron post-training science, medical-o1, Dolci). Every row carries a **source** field recording its origin. Rows failing the gold check are discarded rather than repaired (the answer-gating rule is STaR’s [40]): a wrong trace teaches wrong reasoning. Typical yield is 60–95% depending on source.

Trace genre, and how we measure it. The property that distinguishes a useful trace from a useless one is not correctness but *structure*: whether the text shows reasoning happening (a candidate tried, a flaw noticed, a correction) or merely justifies a conclusion already reached [13, 19]. We score it as

$$S = 0.40b + 0.25v + 0.25p + 0.10\ell$$

where b is the presence of backtracking markers, v of verification, p the fraction that is flowing prose rather than a numbered list, and ℓ a length term. Backtracking dominates because it is the property a write-up never has. S is a *corpus* diagnostic used to decide what to train on; it is not one of the model metrics in §3.

The distinction is not theoretical. A public Greek commonsense-explanation set (ECQA questions translated into Greek with the original human-written justifications kept) scores $S = 0.27$ with 0% of its traces above 0.5; regenerating traces for the *same questions* with the prompt above gives $S = 0.68$ and 97% above 0.5. Our own first corpus was 72% numbered write-ups, which is what sent us looking.

Naming. Arms are named for what actually distinguishes them (the training recipe and the number of reasoning rows) rather than by internal version tags (Table 3):

name	trained on	rows
BASE	nothing (the released model)	—
REASONING	reasoning rows only	59,107
SUBSET	a subset of the same rows	15,607
TWO-PHASE	reasoning rows, then + the direct half	59,107+
ONE-PHASE	one pass over both halves at once	59,107+

Table 3: TWO-PHASE continues the *same adapter* from REASONING onto the hybrid mix; ONE-PHASE never separates the two halves. The three recipes fail in different ways (§7).

SUBSET is a draw of the same reasoning rows, originally selected by a trace-structure score; §4.3 shows that selection performs no better than sampling the same number of rows at random, so we name it by what it is (a subset) rather than by a method the evidence does not support.

Benchmark. 5,156 Greek items: in mathematics, 250 human-translated MGSM-style GSM8K items (ILSP’s `ilsp/mgsm_greek`; MGSM itself [31] contains

no Greek, so these are its 250 GSM8K-test [7] problems human-translated by ILSP, the Meltemi lab) plus our own machine-translated items from the rest of GSM8K test (1,100 of its 1,319 items kept after number-preservation gating, minus 5 duplicated in MGSM = 1,095, 1,345 math in total, §8); Greek HellaSwag [41] and WinoGrande [27] (3,267 commonsense, machine-translated); and a decontaminated ProofWriter-el [34] probe (544 logic, reported as macro-recall). Greedy decoding throughout.

Scoring. Every prompt requests a specific final line. We score *that line* and report **fallback%**, the share of rows where the model never produced it (§8).

Metrics. Accuracy is one of six dimensions we report, not a summary of them: correctness, language fidelity, reasoning budget, termination, reasoning steps and budget overrun. They are defined in full in §3, at the end, because the results are the argument and the definitions are reference; a reader meeting *budget overrun* or *macro-recall* for the first time in §5 can turn there.

Evaluation lanes. The claims in this paper draw on five distinct instruments, and it matters to keep them separate because they measure different things and are run under different conditions. (i) The **Greek reasoning benchmark** above (5,156 items) is the home lane: fidelity, accuracy, fallback, termination and budget behaviour all come from scoring full generated traces on it, in both reasoning (<think>) and direct modes. (ii) The **English control** (1,100 items: the gated English GSM8K originals behind the machine-translated Greek math axis, §8; a separate held-out non-math probe accompanies it) asks whether the model still reasons in English when the question is English; it is scored the same way, and it is what E1 (§10) is read from. (iii) The **Titan-1 suite** (nine Greek plus five English NLU benchmarks, scored by log-likelihood with no generation at all) measures *general* ability for the forgetting question (E2); it cannot see trace language, deliberately: it answers what SFT damaged, not how the model writes. (iv) A **sentence-level switching probe** (E4b method: hand-labelling the language of each sentence in ~150 traces per condition) is what “zero switches per 100 sentences” is counted on; a per-trace ratio cannot see a mid-trace language island. (v) A **Greek NLU benchmark** (18 tasks, 9,751 items: NLI, sentiment, coreference, extractive QA, cloze, machine translation, ...), generated in *direct* mode with no reasoning requested and scored strict/lenient, answers whether the non-reasoning mode still works (E3, §11); its strict-vs-lenient split separates capability loss from format loss the same way **fallback%** does on the reasoning side. Where a number could have come from two instruments, we name the lane next to the number.

3. Metrics

Accuracy is one of six dimensions, not a summary of them. Let an arm a produce, on item i , a reasoning trace $t_a(i)$ of length $w_a(i)$ words and a final answer.

M1: Correctness. Anchored accuracy: the requested answer line is parsed first, free-text matching only if it is absent. Logic is reported as *macro-recall* over its three classes, never accuracy: the class distribution is 891/270/165, so a majority-class answerer scores 67% raw and 33% macro. Accuracy on that axis measures prior-match, not reasoning. Per-class recall shows how literal that is: *every* arm, base included, is near-blind to *Lathos* (False, 14–16% recall across the line) while *Agnotsto* (Unknown) is answered at a prior of 68.8%: the third class works as a decision threshold between two priors, not as a reasoning output. The same mechanism explains why this axis carries the seed swings of §4: a single seed change moved one arm’s *Sosto* (True)-class recall $84.7 \rightarrow 17.3$ (−67 pp) *without* touching the other classes: a True $\rightarrow$ Unknown prior slide on the same data, not a capability change. When a logic number moves, read the class recalls, not the mean.

M2: Language fidelity. $g(t) = |\text{GR}(t)| / (|\text{GR}(t)| + |\text{LA}(t)|)$, the Greek share of alphabetic characters after stripping code and L^AT_EX (otherwise technical traces read as Latin). We report the median g and the fraction of traces with $g \geq 0.9$.

M3: Reasoning budget. Median w_a , and *words per correct answer* $\sum_i w_a(i) / |\{i : \text{correct}\}|$, the quantity a deployment pays for. One confound conditions every word- or token-based number in this paper: Greek costs 2.3–2.5 $\times$ the tokens of English per word *on all three families* (measured fertility: 2.32, 2.39, 2.51 $\times$, see Limitations), so a uniformly-applied **max-new** budget is a roughly 2.4 $\times$ tighter ceiling in Greek. Both M3 and M6 are reported in words, not tokens, by construction: the metric exists precisely to hold the comparison fair across languages; the tokenizer is the hidden hand behind them.

M4: Termination. Share of rows reaching the generation cap without a final answer. Fully objective; no proxy.

M5: Reasoning steps. Count of *intermediate conclusions*: trace sentences asserting a numeric result or explicitly evaluating a named option. Not sentence count, which is $w/12$ in disguise (§9).

M6: Budget overrun. With $f(i)$ the fraction of arms answering i correctly and $\tilde{w}(i)$ the median trace length across arms on i ,

$$\text{OV}_a = \frac{|\{i : f(i) \geq 0.8 \wedge w_a(i) \geq 3\tilde{w}(i)\}|}{|\{i : f(i) \geq 0.8\}|}$$

(overspending on items nearly every model gets right, where difficulty cannot excuse it). Both ingredients are

model-independent: f and $\tilde{w}$ are computed across all arms.

Two diagnostics accompany these rather than scoring quality: **fallback%** (M1’s parse-failure rate, which turns out to measure instruction-following; Figure 1 maps it by recipe and family) and *switch integrity*, the share of think-mode rows returning an empty trace when one was requested.

4. The noise floor

4.1. Why vary the seed

Every claim we had been making has the same shape: *recipe A scored x , recipe B scored y , therefore B is worse*. That inference carries a hidden assumption (that running recipe A twice would return x twice), and we had never tested it. In effect we were comparing interventions against zero, having assumed the noise was zero.

Varying the seed tests exactly that assumption, and it is the control condition rather than an additional experiment. The seed sets the LoRA A -matrix initialisation and the data shuffling order; the corpus, hyperparameters, recipe and step count are identical, the benchmark is fixed and decoding is greedy. Any spread that appears is therefore *training*, and it is the distribution of results produced by doing nothing at all: the null against which every measured effect has to be read.

The closest analogy is calibrating an instrument: before claiming two objects differ in length, one measures the same object twice to learn the ruler’s precision. We had been reporting differences finer than our ruler could resolve.

This is not the error bar usually quoted. The ± 3.4 pp figure that a benchmark of this size implies is *sampling* error: how much the score moves because 1,000 particular items were drawn. It says nothing about how much the *model* moves between runs, and training variance turned out to be roughly twice as large. Papers reporting one number per configuration are accounting for the smaller of the two sources. That training-run variance, not sampling variance, is the operative error bar has been argued before [4, 10, 20]; §17 places this project on that line.

4.2. What it cost, and what it bought

We trained the same reasoning-only configuration three times, varying only the seed (Figure 2: the three runs on the left, every measured effect against the resulting band on the right).

Two runs land near 76 and one collapses to 68.7, so this reads less like symmetric jitter than an occasional failure mode: roughly one run in three lands ~ 8 pp low, taking the logic axis and instruction-following down together.

seed	mean	logic	fallback	trace Greek
42	76.2	56.2	10%	1.00
43	68.7	35.6	41%	1.00
44	76.4	54.0	3%	1.00

Table 4: $sd = 4.4$ pp, range = 7.7 pp. Note the last column.

Either reading supports the same conclusion. For a difference Δ to be detected at the 5% level with 90% power against $\sigma \approx 4.4$ pp, a two-sample comparison needs

$$n \gtrsim 2 \left(\frac{(z_{\alpha/2} + z_{\beta}) \sigma}{\Delta} \right)^2, \quad z_{\alpha/2} = 1.96, z_{\beta} \approx 1.28,$$

seeds per arm: about 11 for our largest effect (6.3 pp) and about 40 for the 3.2 pp one. We could not afford either, and that infeasibility is itself the finding.

Three runs also bound σ itself only loosely: the 95% chi-square interval on $\hat{\sigma} = 4.4$ pp is [2.3, 27.7] pp. The 7.7 pp range is therefore a point estimate from the one configuration and family it was measured on (Qwen); where this paper cites the band against other families or recipes, it is an assumption carried across, not a measurement; the qualitative conclusion (single-run deltas are unreliable) is what transfers, not the number.

The control cost **one additional training run**. It invalidated five conclusions we had already written down (Figure 3 plots every reported effect against the band), and it redirected the paper: the same three seeds leave trace-Greek at 1.00/1.00/1.00 and trace length at 132/148/152 words, so the behavioural dimensions are *stable* under precisely the perturbation that makes accuracy unusable. (One honest caveat: trace-Greek sits at its ceiling, where low variance is partly definitional; the unsaturated trace-length column, varying 132–152 against an accuracy swing of 7.7 pp, is the stronger stability evidence.) We would not have looked for them otherwise. Run this control before the ablations, not after them.

Finding 1. Seed-to-seed variance on a 35B MoE LoRA fine-tune exceeds every data or recipe intervention we tested, replicating the training-variance literature [4, 10, 20] at MoE-LoRA scale¹. Single-run accuracy deltas at this scale are noise. The qualitatively new observation is the asymmetry: *the language and budget dimensions do not move across the same three seeds*: trace-Greek at 1.00/1.00/1.00 (Table 4, last column) and trace length within 132–152 words. That is what lets the rest of the paper measure anything. The asymmetry is not universal: **fallback%** swings 3–41% across the same

¹With one caveat we do not resolve: three of the four conditions are sparse MoE, so whether routing amplifies seed variance relative to a dense model is an open question (Limitations).

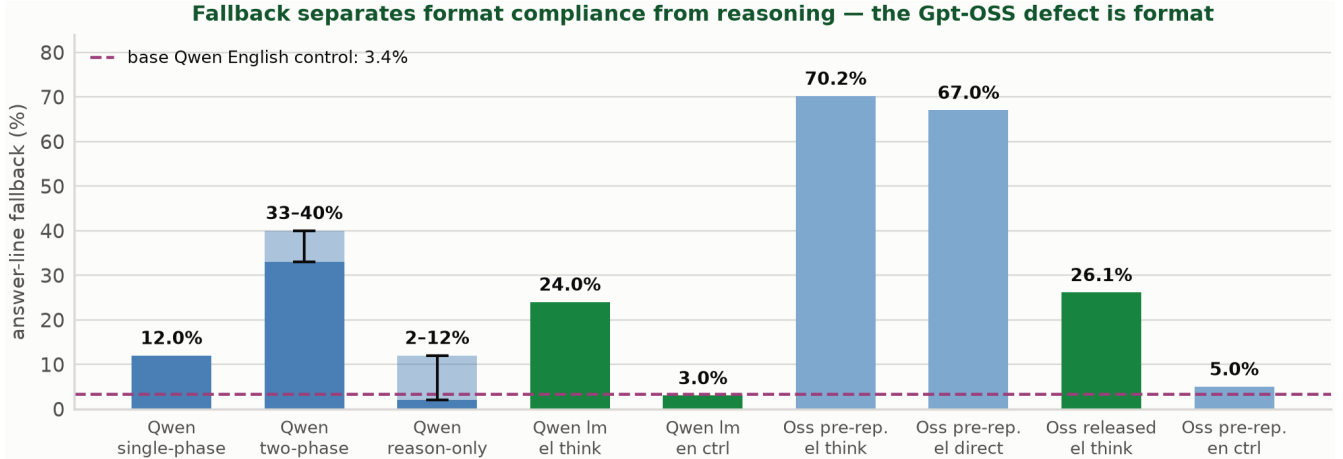


Figure 1: fallback% (the rate at which a model never emits the requested answer line) by recipe and by language-matched checkpoint. It separates “cannot reason” from “will not answer in the requested form”: every recipe failure in the paper is visible here before it is visible anywhere else. The Qwen recipes are the side experiment of §7; the Gpt-OSS language-matched arm’s pre-repair Greek-lane rate (70%) is the family defect at its largest (its traces are present and on-language; the model simply does not close with the line the scorer is told to read), and the released checkpoint’s format-repair dose cuts it to 26% (§11), still the highest of the releases. The dashed reference is a base model on the English control (3.4%), showing the failure is not inherent to the benchmark.

seeds (the same table’s third column), moving *with* accuracy rather than against it (§7).

4.3. Data selection does nothing

The clearest single demonstration is a control we should have run first. SUBSET was built by scoring every reasoning row for trace structure and keeping the top 15,607 (mean structure 0.836). We then trained the identical recipe on 15,607 rows drawn *uniformly at random* from the same pool (mean structure 0.587; Table 5):

phase-1 rows	selection	mean
62,562	all	69.5
15,607	top-by-structure	69.5
15,607	uniform random	69.0

Table 5: Selection buys -0.5 pp (0.31σ); a $4\times$ smaller pool costs nothing. The 62,562 is the reasoning count of the pre-gate *pool* the experiment was run against (62,562 reasoning, 60,214 direct); the 59,107 quoted everywhere else is the post-gate *train* half (59,107 reasoning, 58,985 direct), so the table’s *all* row and the paper’s reasoning-half count differ by the 3,455 rows the decontamination/dedup gate and the val reserve removed.

The pre-registered prediction was that selection would win by more than 2 pp. It is worth being explicit that an ordering across our corpus versions had earlier appeared to show “smaller is better”; that ordering was confounded with which corpus each subset came from, and does not

survive this control. Xia et al. [38] report the same null at scale.

5. What actually changed

5.1. Language fidelity

Given a Greek question, the base models never reason in *Greek*, and mostly not in pure English either. Zero of 1,000 traces reach a Greek character ratio of 0.9 (rule-of-three 95% upper bound 0.3%; the zero holds again on all 5,156 items, Figure 9), the median ratio is 0.33 (English-scaffolded mixed script, only 4.2% purely English), and reading them confirms the English scaffolding (“We need to answer a multiple-choice question in Greek...”). Every fine-tune we trained reasons in Greek on 97.4–98.7% of measured traces (the four SFT releases at 97.4–98.1%, Table 7; the one-directional Qwen arm at 98.7; the RLVR release at 98.27, §15.1).

In our study the property first arrived *overcorrected*: the one-directional fine-tune, instructed explicitly to reason in English, produced a non-English trace on 1000/1000 items: SFT moved something no bare instruction we tried could reach (few-shot untested; Limitations), and moved it too far, since a model that cannot be asked to think in English has lost a capability. The correct target is matching the *question*; retraining on language-matched pairs reaches it (Greek fidelity 97.98% (5,052/5,156), English traces on 100% of the 1,900 paired English items, zero in-question switches either direction), with the full account, including the partial re-opening of

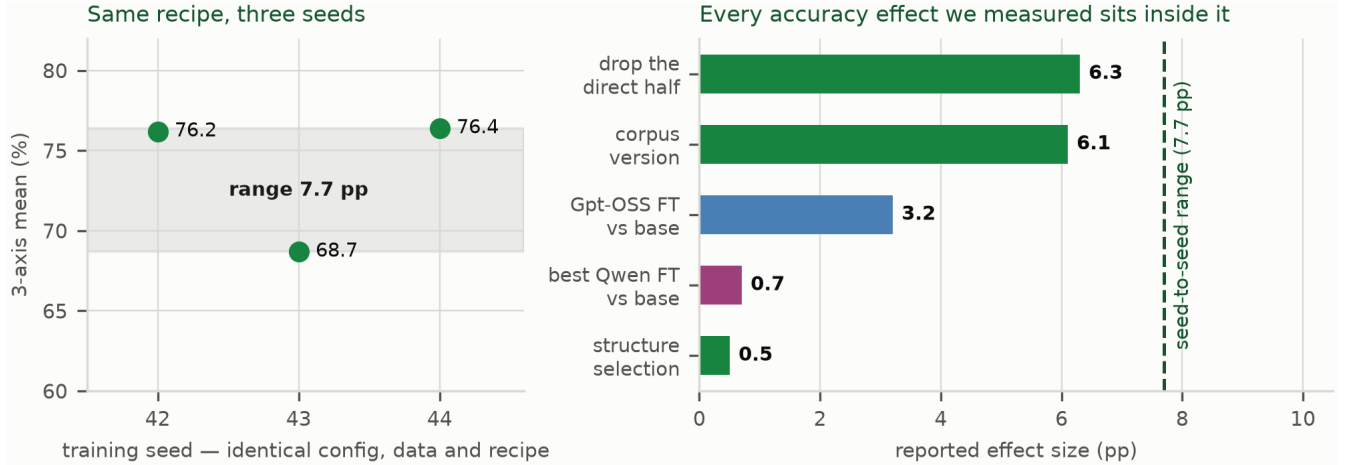


Figure 2: Left: one configuration, three seeds, nothing else changed. Right: every accuracy effect we measured over the project, against that range. All of them fit inside it.

the instructed override on two of the four released checkpoints, in §10 and §10.1.

We test whether the recipe is Qwen-specific: trained on Gpt-OSS-20B, the identical language-matched mix measures 97.79% Greek fidelity pre-repair and 98.14% on the released, format-repaired checkpoint (both within one binomial standard error of the Qwen release), with zero switches in both modes and 100% English-control compliance (Table 12); Nemotron-3.5-Lightning, trained on the same mix, measures 98.06% fidelity, zero switches and 100% English-control compliance under the identical probe (Table 7), with general-ability retention measured against its own base: +1.7 points Greek macro and −1.1 English on the Titan-1 suite (§11, Table 14); it is released as a third matched checkpoint, not a full replication of the two. The Gpt-OSS arm does show one family-specific cost: its answer-channel leak is 9.1% pre-repair and 10.3% on the released checkpoint, against Qwen’s 3.5%, so the trace-register boundary that §3 measures holds at different tightness per family.

The Gpt-OSS arm also separates the two failure modes this paper keeps apart. Before its format repair, its Greek-lane anchored accuracy reads 56.2, with 70.2% of rows never emitting the requested answer line, so the score is almost entirely a *format* floor, not a reasoning one (§3): the reasoning is present and on-language; the model does not close with the line the scorer reads. Both language-matched checkpoints pay a format tax: Qwen’s **fallback%** is 24 on the Greek lane (and 3 on the English lane); the released checkpoint answers in the requested form on three of four Greek items. Gpt-OSS’s is 70/67 (think/direct) before the repair, with 5 on English; the released, format-repaired checkpoint pays 26 on the Greek think lane (§11), still the highest of the releases. The fidelity result above is independent of this metric: it is

measured on the trace, where the released checkpoints are on-language at 98.0–98.1%.

Reading the same releases per domain sharpens that format-versus-capability reading (Table 6). The defect the Gpt-OSS arm pays is not uniform: before repair its fallback was **35%** on math against **83%** on commonsense and **77%** on logic: the loss lives in the open-form domains, while math’s short numeric answer is format-robust. That is the shape a format defect takes, not the shape a capability loss takes. Repair rescues the worst domain first (commonsense 83→26% and logic 77→51% fallback, math 35→17%), and accuracy follows. Two smaller readings. Logic is the hardest axis for *every* arm (base macro-recall 32–43% across the two base rows), so its low scores are a floor, not a regression. And a correction this revision owes the reader: an earlier draft read the Qwen release as *gaining* on commonsense (63.2→75.8), but the 63.2 comparator in Table 6 is the *Gpt-OSS* base; against its own base’s 82.4 on the same lane, the Qwen release is −6.6 on commonsense, and no fine-tune beats its own base on any axis of Table 6. The Qwen release’s logic fallback is 46%, and the repaired Gpt-OSS release’s higher still (51%).

One more observational claim we believed and then had to withdraw belongs here, because it is the question every reader asks next: *is Greek reasoning worse reasoning?* An early run of the fine-tune on Greek items scored −17.5 pp against the same items in English, which reads seductively as “the model thinks better in English.” The controlled version separates the two: run the *same* items, force the *same* answer language, vary only the trace language (§8’s answer-format control is the same trick in miniature). The language effect collapses to +1.4 pp (0.65 σ): statistically nothing. The −17.5 pp was selection (a different, easier question mix in the English lane)

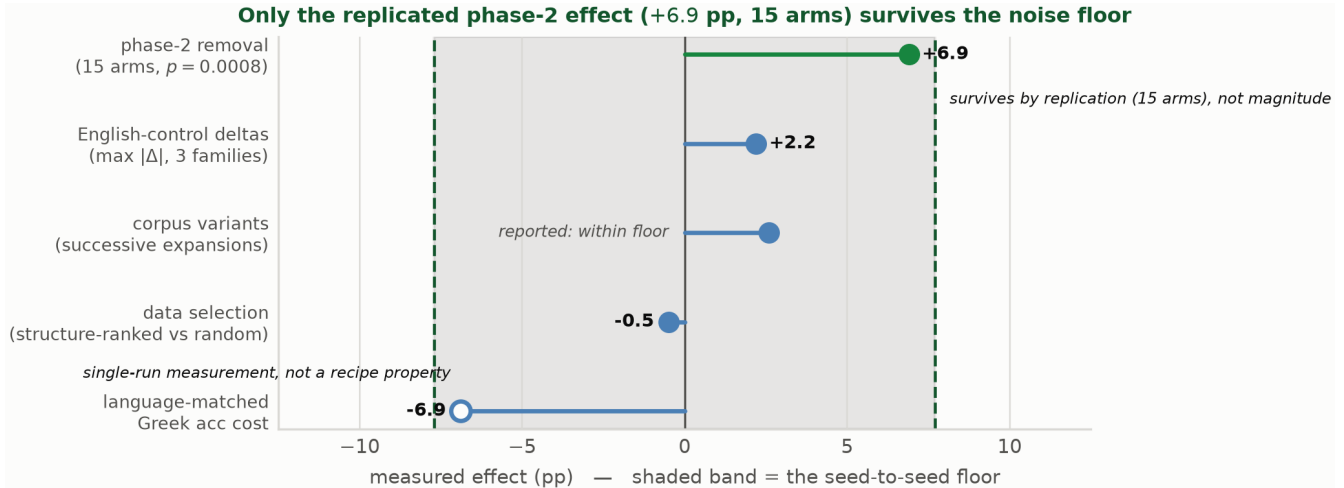


Figure 3: Every accuracy effect we measured, against the 7.7-point seed-to-seed band (shaded). Attribute-level effects (data selection and corpus versions) sit inside the band we would have been tempted to narrate. The only effect that survives is the replicated phase-2 comparison of §7 (+6.9 pp, 15 arms, permutation $p = 0.0008$): it is inside the band geometrically, but it is estimated across 15 independently trained arms rather than read off a single run, which is precisely the difference the floor teaches. The language-matched accuracy cost (−6.9 pp) is drawn as an open (unfilled) marker on purpose: it is a single-run measurement inside the floor, disclosed rather than interpreted, and not a recipe property. The equal magnitudes of the +6.9 and the −6.9 are a coincidence: different lanes, different arms.

expressing itself as a language effect, and it is the largest retelling-of-a-difference the project ever wrote into a slide before the control.

5.2. Reasoning budget: words fall everywhere it matters, tokens change sign by family

At identical item-pooled accuracy (72.9 vs 72.9 on the same 1,000-item probe; the three-axis strict mean there is 77.2), the base spends a median 1,010 words per trace and 1,454 words per *correct* answer; the reasoning-only fine-tune spends 150 and 239 (Figure 4 sets this beside the flat accuracy panel). In *words*, our length unit throughout (§3), the base-to-fine-tune ratio is $3.7\times$ for the Qwen language-matched release and $1.5\times$ for the NemotronH fine-tune (the earlier one-directional arms are more favourable still: $4.5\text{--}5.1\times$ Qwen, $1.4\times$ Gpt-OSS). But words are not what a deployment pays, and Greek costs $2.3\text{--}2.5\times$ the tokens per word that the bases’ English traces do (the measured fertility ratios of §3). Re-tokenizing every trace on the full benchmark with each family’s own tokenizer, the saving changes sign by family: the Qwen release spends $3.0\times$ fewer tokens than its base (median 586 vs 1,788), the NemotronH fine-tune sits at **parity** (638 vs 681, $1.07\times$), and the Gpt-OSS release spends $1.6\times$ more (640 vs 396): its traces are already slightly longer in words than its terse base’s, and the fertility gap widens the difference. We therefore split the claim: the reduction in reasoning *effort* (words) is real wherever the recipe shortens traces, but the *serving-cost*

saving survives translation into tokens only on Qwen; on Gpt-OSS the Greek fine-tune is more expensive to serve than its English-reasoning base (Figure 5).

We state this as a marginal claim only. Conditioning on trace length, the base is at least as accurate as the fine-tune, and *within every arm longer traces are less accurate* (base 82.5% under 400 words vs 72.3% over). Length marks items a model is struggling with, not effort that pays off. The fine-tune’s advantage is that it does not enter the long unproductive regime, not that its tokens are worth more. Both stratifications are endogenous, so no causal claim about length is available from this design.

5.3. Budget discipline

Measuring degenerate looping directly proved impossible: looping is what *makes* a trace long, so every repetition metric we tried was a length proxy (§8). The measurable question is whether a model spends far more than a given item requires. Let $f(i)$ be the fraction of arms answering item i correctly and $\tilde{w}(i)$ the median trace length across arms on that item; an arm overruns on i when $w(i) \geq 3\tilde{w}(i)$. Conditioning on the item supplies the control a raw rate lacks.

The base overruns on **98.0%** of easy items, against 0.2–12% for every fine-tune (Figure 6). And its rate is identical on easy and hard items (98.0 vs 98.4): it has no representation of question difficulty. Fine-tuned arms range $1.9\text{--}60\times$ between the two in *direct* mode, spending more only when the item is harder; think-mode ratios are

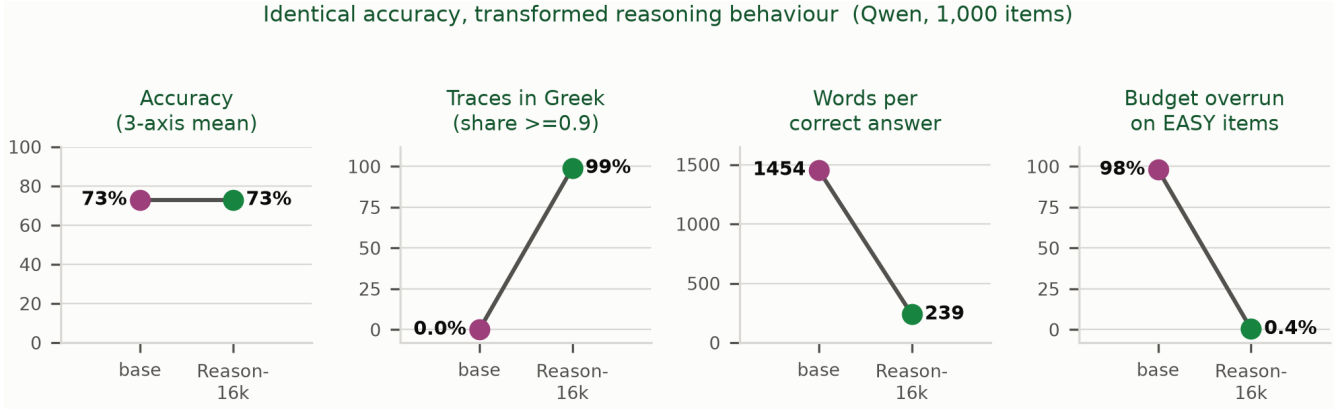


Figure 4: The same two checkpoints on four dimensions, the base and the structure-selected reasoning-only arm (SUBSET, the “Reason-16k” of Figure 11), on the 1,000-item probe lane. Only the first is accuracy.

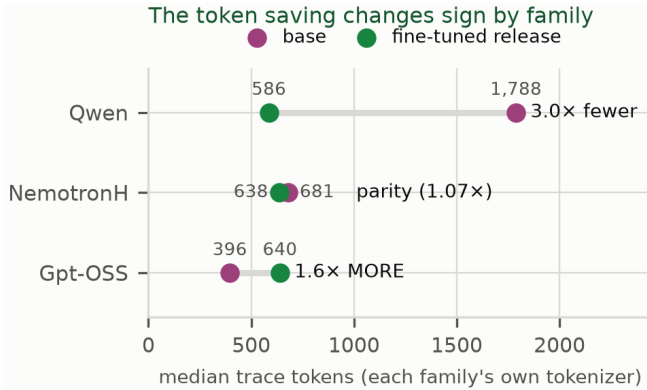


Figure 5: Median trace length in *tokens* (each family’s own tokenizer, 5,156 matched items), base against the fine-tuned release. The word-level shortening must repay Greek’s 2.3–2.5 $\times$ token fertility before it becomes a serving saving: Qwen’s does, NemotronH’s breaks even, Gpt-OSS’s does not.

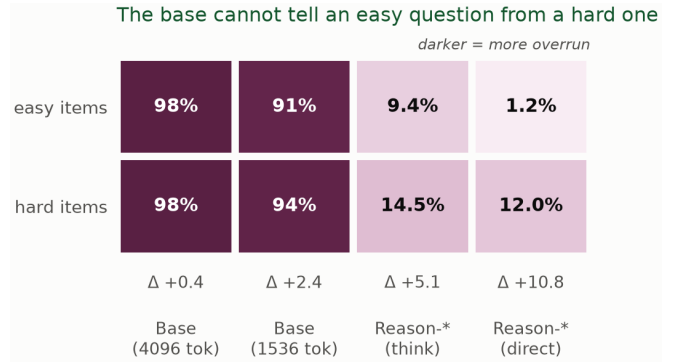


Figure 6: Overrun = trace $\geq 3\times$ the median trace *for that item* across arms. Easy = at least 80% of arms answer correctly. Cells pool the arms of each column; hard-item think-mode rates are compressed by generation-cap truncation (§5), so the easy row carries the section’s claims.

compressed by generation-cap truncation on hard items, which is why the easy-item *rate*, unaffected by the cap, is the figure this section reports.

Finding 2. We find that at equal accuracy the fine-tune reasons in Greek rather than English and adapts its budget to question difficulty, but the token cost of doing so is family-dependent in *sign*: 3 $\times$ fewer tokens on Qwen, parity on NemotronH, 1.6 $\times$ more on Gpt-OSS, because Greek’s 2.3–2.5 $\times$ token fertility must be repaid by shortening traces (§5). None of these appear in a benchmark mean, and none move across seeds.

6. Losses that looked like forgetting

If SFT installs Greek reasoning, what does it remove? The answer sorts into three buckets, two of which are measured here, and the third is the one readers most often conflate with forgetting: the language lock. That is

a capability keyed to the training direction, not a deletion, and its full treatment lives with the pre-registered E1 design in §10, including Table 10. For general ability itself, the forgetting question is answered on the Titan-1 suite in that same section (Table 13): flat in both languages on two families, with a -3.2 -point Greek residual on the format-repaired Gpt-OSS release. This section carries the two remaining appearances of forgetting, both of which dissolve on contact with the right control.

(a) **Format compliance degrades, and the recipe decides how much.** Failure to emit the requested answer line rises from 12% (single-phase) to 33–40% (two-phase) and stays low 2–12% (reasoning-only, with seed noise widening that to 2–41% as disclosed in Table 8). The forgotten capability is format compliance, and §7 localises it to the second phase. It is a recipe choice with a measured mechanism, not forgetting: the language-matched checkpoints of §5 pay it too, at 24% (Qwen) and 26% (the format-repaired Gpt-OSS release; 70% be-

arm	anchored accuracy			fallback %		
	math ↑ higher is better	cs ↑ higher is better	logic ↑ higher is better	math ↓ lower is better	cs ↓ lower is better	logic ↓ lower is better
Qwen base (Greek think)	92.9	82.4	42.6	3.3	0.5	0.2
Sopheia-Qwen3.6-v1	83.3	75.8	37.3	18.2	22.9	46.3
Gpt-OSS base (Greek think)	90.7	63.2	32.2	2.4	5.2	0.7
<i>before repair (reference)</i>	74.6	53.7	25.2	34.9	83.5	77.4
Sopheia-OSS-v1 (repaired)	78.6	53.3	26.5	17.2	25.7	50.9
NemotronH base (Greek think)	86.6	49.2	26.8	8.3	6.2	6.6
Sopheia-Nemo-3-Nano-v1	71.5	39.7	9.8	1.7	1.8	39.9
Nemotron-3.5 base (Greek think)	76.9	46.3	25.0	1.4	1.9	4.4
Sopheia-Nemo-3.5-Lightning-v1	75.6	40.9	23.8	16.1	14.7	5.3

Table 6: Per-domain results on the Greek think lane; here and in every table, ↑ marks metrics where higher is better and ↓ where lower is better (math = the two math slices, commonsense = HellaSwag+Winogrande, logic = ProofWriter macro-recall). Rows are grouped by family, each release indented under its own base, so every legitimate accuracy delta is a within-block comparison and the cross-base reading (the one an earlier draft of this paper fell for) is structurally discouraged. Every release now carries its own base row: the NemotronH base dump is the same vintage as the Qwen base dump, and the Nemotron-3.5 base lane, absent until this revision, was generated with the identical lane settings and scored with the same scorer as every other row. The same base is also measured on the NLU retention suite (Table 14, Greek macro 57.5), a direct-mode instrument whose numbers are not comparable to this think-lane table. The Nano release also appears in the retention suite (Table 13), and its fallback shape is the mirror of Gpt-OSS’s: negligible on math and commonsense (1.7/1.8%) but 39.9% on logic, so its 9.8 logic macro-recall is substantially a format floor on the one domain where its answer form breaks. Accuracy and fallback move together: every release keeps the base’s domain ordering, and the Sopheia-OSS format-repair dose pulls the commonsense fallback back toward math’s level (83.5→25.7) while cutting logic’s by a third (77.4→50.9). The pre-repair row is reference only; the release is the repaired arm. Conditioning on rows that emit the requested answer line (excluding the fallback-scored path) shrinks the apparent deficits sharply: the Qwen release reads 95.8/78.8/39.5 against its base’s 96.0/82.8/42.7, and the same condition makes Gpt-OSS pre-repair 74.6 → 92.2 / 53.7 → 60.2 / 25.2 → 36.5 and Sopheia-OSS-v1 78.6 → 94.4 / 53.3 → 58.2 / 26.5 → 30.0 (math/commonsense/logic), so a substantial share of every raw gap is format compliance, not reasoning.

fore its repair) on the Greek lane.

(b) The commonsense loss is not forgetting. A −3.0 pp commonsense deficit appeared in 12 of 12 fine-tuned arms and looked like textbook catastrophic forgetting. Re-asking the same items in a constrained answer format reverses it to +1.7 pp (§8). The capability was never lost; only its expression under long-form generation changed. We report this because the negative result is the more useful one: an apparent forgetting effect that survives twelve arms can still be an artifact of how the answer was elicited.

Register control and grammaticality: not lost, and mostly gained. Two judge-based probes close the forgetting picture from the fluency side, run identically on every release and its own base (an LLM judge at temperature 0; $n=46$ register items and $n=58$ morphosyntax items, small enough that we report counts, not percentages). *Register control* (produce the requested formal or informal register) is never lost: the two weakest bases gain it (Nemotron-3.5, 37/46 → 43/46; NemotronH, 40/46 → 44/45), Gpt-OSS is flat (38/45 → 39/46), and Qwen dips within small- n noise (45/46 → 42/46). *Grammatical correctness* (agreement, clitics, and related mor-

phosyntax) improves on every family, and improves most where the base is weakest: NemotronH 13/58 → 27/57, Nemotron-3.5 12/58 → 29/58, Gpt-OSS 32/57 → 40/58, Qwen 41/58 → 42/58. One instrument note in this section’s own spirit: the first-pass Gpt-OSS numbers were an artifact of a response-splitting bug that prefixed a stray channel marker to otherwise correct answers; the judge failed them for the prefix, the splitter was fixed, and the fine-tune’s dumps were re-judged (the corrected numbers are the ones above, and the direction of the finding reversed from loss to gain).

Finding 3. We find that almost nothing the fine-tune appeared to forget was actually forgotten. The language lock is conditional on training direction, not a deletion (full treatment: §11); general ability is flat on two families, with a −3.2-point Greek residual on the third after its format repair (Table 13); the commonsense regression was a format artifact; and the one genuine loss (answer-format compliance) is recipe-localised and disclosed for the release checkpoints.

On the fluency side the sign flips outright: register control is retained or gained on every family, and judged

arm	Greek fid. (%)↑	EN ctrl (%)↑	switches /100 sent.↓	leak (%)↓
Qwen base	0.0	100	16.4	15.6
Qwen lang-matched	98.0	100	0.0	3.5
Qwen one-dir. [†]	98.7	99.3 [§]	0.0	1.5
Gpt-OSS base	0.0	100	9.7	0.5
Gpt-OSS lang-matched	98.1	100 [‡]	0.0	10.3
Nemotron-3.5 lang-matched	98.1	100	0.0	5.2
NemotronH lang-matched	97.4	100	0.0	0.0
NemotronH one-dir. [†]	98.4	100	0.0	11.3

Table 7: Language-matching across the three families: all four released checkpoints, the two measurable bases, and the one-directional reference arm. *Greek fid.*: Greek-trace fidelity on the Greek benchmark; both bases sit at exactly 0.0 (0/5,156 traces reach ratio 0.9), and all four language-matched checkpoints land at the same level (97.4–98.1%). *EN ctrl*: English-control compliance, 100% for every arm. *Switches*: in-question language switches per 100 sentences on the Greek lane; the bases switch 16.4 (Qwen) and 9.7 (Gpt-OSS) times, every fine-tuned arm (the one-directional reference included) zero. The base fidelity and switch cells are one instrument run: both base dumps rescored with the current scorer on the same day (an earlier draft printed 19.0 for the Qwen base from an earlier dump vintage). *Leak*: answer-channel leak, the one place families differ (10.3% Sophea-OSS-v1, 5.2% Lightning, 3.5% Qwen, 0.0% Nano, the only arm at zero; base leak is measured on the English control, fine-tuned leak on the Greek think lane). [†]Arms of the earlier one-directional recipe, shown for reference (a different recipe, not releases); the Qwen one-directional row is the recipe-evolution comparison: it locks against explicit language *instructions* (Table 10), not against English questions. [§]Measured on the 1,900-item paired probe (99.3% English, 0.0% Greek, 0.7% too short to score); this arm was never run on the 1,100-item control the other EN-ctrl cells use. [‡]Measured on the pre-repair language-matched arm: the repair dose adds 2% trace-less answer-format rows and does not touch trace language, but the released arm’s own English control has not been generatively re-run. The recipe generalises to every family tried; the leak’s size does not.

grammaticality improves on all four, most where the base is weakest. What a benchmark calls forgetting is, in this project, almost always something else. Flat macros are also partly the expected property of a LoRA adapter rather than of our data [3]; §11 carries that scoping in full, and the Limitations section keeps it LoRA-conditional on purpose.

7. A side experiment: three recipes, and what the second phase costs

This section stands apart from the paper’s main line. The question it answers (*how much of the training corpus should be non-reasoning, and in what order should the halves be shown*) was settled for our release long before

the language-matching result existed, and none of the main claims depends on it. It stays in the paper for two reasons: it supplies the mechanism for our answer-format failure mode (§6b), and it contains the one accuracy effect in the project that survived the noise floor, a result we would have to explain the absence of if we omitted it. Read it as a controlled detour, not as a load-bearing section.

7.1. The three recipes, and the trade they make

The corpus has two halves: reasoning rows carrying explicit traces, and ~59k non-reasoning conversational rows. Three ways to use them, and each fails differently.

recipe	mean	empty	fallback
ONE-PHASE	66.1	23.6%	12%
TWO-PHASE	64.4–69.9	0.0–1.3%	33–40%
reasoning-only	73.6	0.0%	2–12%

Table 8: *single-phase* = one pass over both halves; *two-phase* = reasoning-only then hybrid; *reasoning-only* = the reasoning half alone. Empty-trace rate is measured in think mode, **fallback%** in direct mode; means are over all arms of each recipe. Fallback ranges are over the individual arms of each recipe (2–12% across the three reasoning-only arms; 33–40% across the two-phase arms); the three seed replicates of §4 widen it further (3–41% across seeds 42–44, all three of which are among the six phase-1-only arms above), so the fuller picture is 2–41% when seed noise is admitted.

In our runs, single-phase SFT collapsed the reasoning switch. Trained in one pass over a corpus that is half non-reasoning, the model learns to answer directly regardless of the flag: asked to think, it returns an *empty* trace on 23.6% of items. This is the cleanest causal result in the line and it is coverage-independent: it counts empty traces, not answers.

Two-phase fixes the switch and breaks something else. Training reasoning-only first, then continuing the *same adapter* on the hybrid mix, drives empty traces to 0–1.3%. But answer-format compliance degrades from 12% to 33–40% fallback, and accuracy does not improve. The second phase repairs the mode switch at the cost of instruction-following.

Phase-1-only avoids both, and gives up the direct mode. Never showing the model the non-reasoning half yields the best accuracy *and* the best compliance. At the time this experiment ran, the cost was architectural rather than measured: such a model has only ever been trained to reason, so it has no separately trained direct mode to collapse. E3 (§11) later measured that cost directly, and found none detectable on direct-mode NLU.

7.2. The one accuracy effect that survives the floor

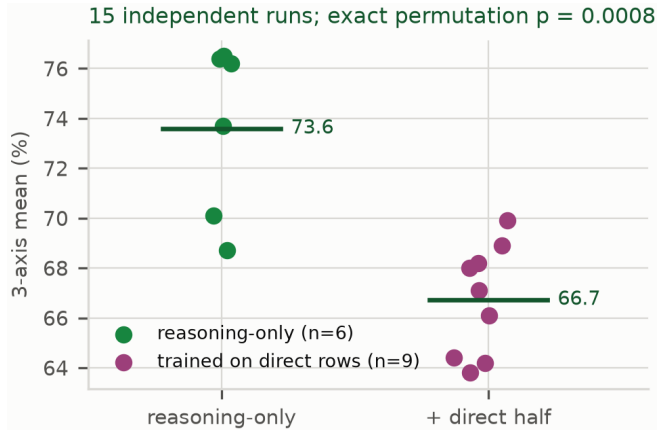


Figure 7: Each point is an independently trained arm. Corpus version varies within both groups.

A single reasoning-only versus two-phase comparison (+6.3 pp) sits inside the noise floor and cannot be claimed. But the distinction separates *fifteen* independently trained arms (Figure 7), and across runs that is the correct unit of analysis: reasoning-only averages 73.6 ($n=6$) against 66.7 ($n=9$), with 52 of 54 pairwise comparisons favouring it and an exact permutation $p = 0.0008$. On the decontaminated benchmark the effect is +5.3 pp ($p = 0.0164$), and on the 5,156-item benchmark reasoning-only wins *every axis* (+5.4 pp mean).

One confound this design cannot fully exclude: the arms are the project’s historical runs, and corpus version varies within both groups rather than being balanced across them. Stratifying the permutation by corpus version leaves only three strata with members on both sides and 16 valid permutations in total: the observed split ranks second of the sixteen ($p = 0.125$, attainable floor 0.0625), so the stratified test is supportive but has almost no resolution. The $p = 0.0008$ figure assumes exchangeability across corpus versions; the fact that four versions appear on both sides argues for that assumption qualitatively, but a designed, version-balanced replication is what would settle it.

The mechanism is visible in **fallback%**: reasoning-only arms fail to emit the requested answer line on 2–12% of items, two-phase arms on 33–40%. The wider range in the seed replicates reasoning-only alone (3–41% across the three seeds of §4) is seed noise, not a recipe effect; an out-of-family arm on a much bigger corpus without phase 2 sits between, at 24%. Training on ~59k non-reasoning rows degrades instruction-following, and part of what looked like a reasoning regression is a model that still reasons but no longer answers in the requested form.

The three seed replicates of §4 are themselves three of

the six reasoning-only arms, so the seed spread is represented inside the aggregate rather than hidden from it. This is **Qwen only**. For Gpt-OSS the ordering reverses (62.5 vs 68.1) with one run per condition, so it is unestablished there. Both statements belong in the record: a single A/B is uninterpretable at this variance, *and* the aggregate is significant.

The side experiment’s own finding. Removing the non-reasoning half of the corpus is worth +6.9 pp across 15 runs ($p = 0.0008$): the only accuracy effect large and replicated enough to survive the noise floor, and the only one whose mechanism we can point at. It bears on the main line in exactly one place: a deployment that never needs a direct mode should train reasoning-only (§14), and that is a deployment choice, not a property of the language results.

8. Six ways our instruments lied

The first four of the following each produced a plausible result that we believed, wrote up, and withdrew; the fifth was a small correction caught by audit, and the sixth was caught in-flight, before its wrong numbers reached a table. Figure 8 draws all four before-and-after pairs.

8.1. A scorer that ignores the requested answer line

Our prompts request “... write on a new line: The answer is <letter>”. Our scorer did not privilege that line; it scanned the whole response and let the last option-mention win. A model that states its answer and then explains why the others are wrong is therefore scored on its explanation. The bias is one-directional (verbose arms are penalised, terse arms are not), and on one probe it produced a +29.8 pp (18σ) artifact favouring the fine-tune. Anchoring extraction to the requested line removes it. *Any benchmark that requests an answer format must score that format first and report how often it was absent.*

8.2. A capability gap that is an artifact of the answer format

Our headline negative result was a −3.0 pp commonsense deficit, present in 12 of 12 fine-tuned arms. Re-asking the same 300 items with a letter-only prompt reverses it to +1.7 pp. The gap exists only when models reason at length, and the claim as stated was withdrawn. Format must also be chosen per axis: letter-only is valid for commonsense and invalid for logic, where it drops the base to 29.6% against a 33% chance baseline.

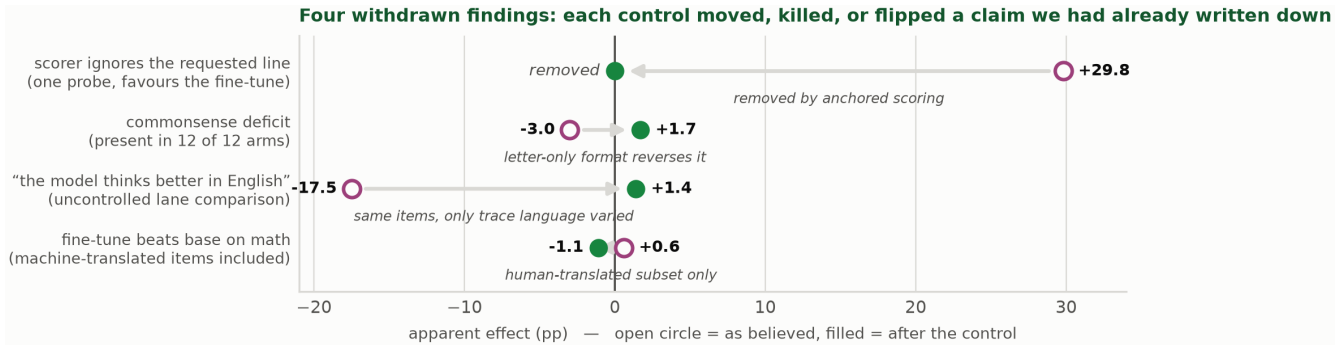


Figure 8: The four withdrawn findings of this section, drawn: the effect as first believed (open circle) against the same quantity after its control (filled). Three flip sign; the un-anchored scorer artifact (+29.8 pp, 18σ , one probe) collapses entirely: its endpoint is drawn at zero because the prose reports the artifact removed by anchored scoring, not as a measured residual. Rows are different instruments and lanes, so magnitudes compare within a row, not across rows; the other pairs’ numbers appear in this section’s prose. The section’s fifth failure (§8.5) is a scorer defect rather than an effect-size withdrawal, so it has no before/after pair to draw.

8.3. Contamination that a standard check does not find

Our logic axis was **38.9%** contaminated: 175 of 450 items shared a 13-gram with the training pool, while math and commonsense were clean at 0%. (The 450 here is the *original* logic probe, a hard-subset sampling of the ProofWriter-el pool; the 544-item axis in §2 is a different instrument on the same pool; it is the full draw after removing the 782 contaminated items. The 38.9% figure is therefore the probe’s contamination rate, and the decontamination cost that left the full pool at 544.) The cause is structural (the benchmark and the training slice were drawn from the same ProofWriter pool), and the original build script verified contamination for the other sources but not this one, because that pool was assumed eval-only. Removing the items moved individual arms by 0.7 to 22.2 pp, i.e. *unevenly*, which is what differential memorisation looks like. (The *n*-gram check used throughout is a surface-match method of the kind Ravaut et al. [25] survey; its known blind spot is paraphrase, which matters for translated suites, and the Limitations section records that we did not test beyond it.)

The check also has to run in *both* languages once the training corpus does: the language-matched corpus carries the English originals of its training questions, so we ran the same 13-gram check between its English half (21,383–24,509 rows across the two corpus versions) and both English evaluation sets (the 1,100-item GSM8K-EN control and the paired English probe). **Zero rows collide.**

8.4. Translating a benchmark with the model family you trained on

To scale the math axis we machine-translated GSM8K’s test split into Greek, keeping 250 human-translated items (ILSP’s MGSM-el set, §2) as a validity check. The check failed for exactly one arm (Table 9):

arm	human	machine	Δ
BASE	95.6	90.7	-4.9 (-3.14σ)
SUBSET	94.0	95.1	+1.1
REASONING	93.6	95.3	+1.7
TWO-PHASE	87.6	90.4	+2.8

Table 9: The base is ~ 5 pp worse on our translations; every fine-tune is slightly better.

Our training traces and these translations both came from the same commercial model family (§2); the translation model is the same one that generated the mathematics traces. So the fine-tunes are adapted to its Greek register and the base is not. Including the machine-translated items makes the fine-tune appear to beat base by 0.6 pp; on the human-translated subset the base leads by 1.1 pp, consistent with an independent benchmark. *If you build a low-resource benchmark by LLM translation and train on data from the same family, your fine-tune gains an advantage no contamination check will find: the items are novel, only the register is shared.* A 250-item human-translated control costs almost nothing and caught a 3.1σ artifact.

8.5. A scorer that cannot read Greek numbers

The fifth failure is in the scorer that produced every accuracy number in this paper, and we found it only when au-

ditioning the same code for use as a reinforcement-learning reward. Greek, like most of Europe, writes 17.500 for seventeen-thousand-five-hundred and 3,5 for three-point-five: the separators are the reverse of the English convention. Our extractor strips commas and calls a float parse, so a model answering 114.200 is read as 114.2 and marked wrong.

On the released Qwen checkpoint’s math axis, six answers are written in Greek thousands form and **five of them are scored wrong while being right** (114,200, 17,500, 43,200, 7,300, 1,800). The mirror case exists and is worse in kind: a decimal comma such as 3,5 has its separator stripped and becomes 35, which can score a wrong answer as correct.

The size of the effect on this paper is small: 5 of 1,100 anchored math rows, 0.45 pp on one axis, inside every noise floor we report, and no number in this paper changes. We report it because the *shape* of the error is the point and generalises past us: the defect is invisible to an English-language test suite, it is systematic rather than random, and it penalises exactly the answers written in the target language’s own convention. A benchmark built for a non-English language needs its numeric normalisation tested in that language’s conventions, in both directions, or it will quietly score fluency as failure. *Locale is part of the instrument.*

8.6. A default that silently changed which lane we measured

The sixth entry was caught while evaluating the RLVR round (§15), by the control this section keeps recommending: a same-day baseline. The fresh baseline read 1.1% fallback where the frozen number was 24%, too large for noise in either direction. Rescoring the *original* dump with the current scorer reproduced 24.1% exactly, acquitting the scorer; the generations themselves differed. The cause was a generation-harness default: without an explicit flag, the chat template rendered with the reasoning trace *disabled*, so four GPUs spent four hours generating the direct lane under a filename that said think lane. Nothing crashed, nothing warned; the numbers were internally consistent and wrong. Every mislabeled dump was discarded, the flag is now part of the instrument definition next to the locale rule above, and the incident is why Table 16 states its *before* column’s provenance explicitly. The general form: *a default that selects which condition you measure is not a default, it is a hidden factor*, and only a same-day regeneration of a known quantity will catch it, because every downstream number is plausible.

Finding 4. Five controls, five withdrawn or corrected findings, and a sixth failure caught in-flight by the same discipline (§8.6). Each original claim was observational or single-condition, and each agreed with what we expected, which is why it survived review until the control was run. The fifth was found in the scorer itself, and only because we re-read it for a different purpose (§8.5): an instrument can be wrong for a whole language and still look right on every English test.

9. A metric needs a control, not a normalisation

We tried three definitions of degenerate looping. All were length proxies, with $|\text{corr}(\text{words}, \text{metric})|$ of 0.44–0.85 for a per-token rate and 0.86–0.95 for a fixed-window rate on the arms whose traces are shorter than the window. Normalising by length does not fix a metric that length causes; the fix is a comparison in which length is held constant by design, which is what §5 does by conditioning on the item.

We therefore gate every behavioural metric: any candidate correlating $|r| \geq 0.6$ with trace length is redefined or dropped, never reported with a caveat. The gate removed three of our seven original dimensions, including a “the fine-tune loops 29× less” claim that was pure length artifact.

The measured consequence is worth stating once, because it is asymmetric in a way no length proxy captures. Applying a repetition penalty suppresses the degenerate loops in both the base and the fine-tune: at *zero* cost to the base (−0.4 pp within the noise floor) and at a −9.7 pp cost to the fine-tune. The base loops *when it has nothing to say*; the fine-tune loops *as part of how it reasons*: its repeated spans are load-bearing. Removing them does not clean up its style; it removes part of its argument. The asymmetric price of the same constraint is the sharpest evidence we have that degeneration means something different in the two models.

10. Fixing the lock: train on matched language pairs

The lock of §5 is not a property of Greek reasoning SFT as such. It is a property of *training the reasoning language in one direction only*, and it lives in the instruction channel, not in the default: the one-directional arms still reason in English when the question is English (Table 12), and the locked Qwen arm ignores an explicit instruction to switch (0/1,000), a severity that is family-dependent under the same recipe (the Gpt-OSS one-directional arm stays 95% steerable, §13). A checkpoint re-trained on **language-matched pairs** keeps the question-following default on both lanes for every family we trained, and re-

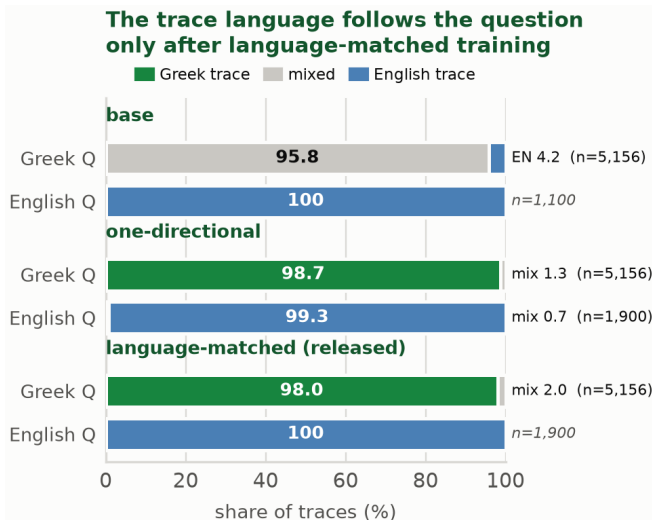


Figure 9: Trace-language composition per lane, Qwen line: the share of traces that are Greek (character ratio ≥ 0.9), mixed, or English (≤ 0.1), on the same instrument as Table 7’s fidelity column. Greek-question rows are the full 5,156-item benchmark; English-question rows are the 1,100-item control (base) and the 1,900-item probe (matched). Two honesty notes drawn rather than hidden: the base’s Greek-question traces are *not* mostly pure English: only 4.2% are; the mass is mixed script with English scaffolding (median ratio 0.33, §5); and the one-directional arm’s Greek-question lane decomposes as 98.7 Greek / 1.3 mixed / 0.0 English on the same instrument ($n=5,156$, uninstructed); its English-question lane, generated later on the same 1,900-item probe, measures 99.3% English (Table 7). Uninstructed defaults only; the instructed override is Table 10.

opens the instructed override on *two of the four* released checkpoints (44.8–62.5% compliance on Qwen and Gpt-OSS; 0.0% on both Nemotron arms, §10.1). Figure 9 draws the default as trace-language composition per lane. This section reports that result self-contained: it is the answer to the first of three questions we pre-registered (the other two are answered in §11, each by an evaluation whose design was fixed before the result), and it changed the paper’s direction.

E1: was English reasoning lost, or merely not elicited? Answered in part: not the capability, and not the recipe alone; the *direction* of training, with a family-dependent severity we cannot yet explain (the Gpt-OSS arm under the same one-directional recipe stays 95% steerable, §13). The one-directional fine-tune *looks* completely locked, and it stays locked when you look harder. Instructed explicitly, on the same 1,000 items:

No prompt we tried recovers the English, so read on this arm alone, the capability looks gone. It is not gone; the cause is the one-directional recipe, as this section

model, instruction	English traces	Greek traces
base, “reason in English”	100.0%	0.0%
base, “reason in Greek”	72.0%	0.0%
fine-tune, “reason in English”	0.0%	98.7%
fine-tune, “reason in Greek”	0.0%	98.7%
matched ft, “reason in English”	44.8%	24.8%
matched ft, “reason in Greek”	0.0%	98.5%

Table 10: The one-directional fine-tune’s traces are identical under both instructions, and it does not matter how the instruction is phrased. Rows classified as neither/mixed scripts (28% of the base under the Greek instruction) are omitted, so rows need not sum to 100. The *matched ft* rows, measured after the language-matched re-training of §10, show the override partially returning; its mixed share under the English instruction is 30.4%.

opened. A checkpoint re-trained on **language-matched pairs** (each Greek problem and trace kept beside its English original and trace, gated so a trace that drifts languages never enters training) reasons in English on **100%** of 1,900 English questions (median Greek-character ratio 0.000, held across mathematics, science, medical and general domains), while the English-accuracy cost is -2.2 pp (-2.46σ), half of which is answer-format fallback rather than wrong reasoning. The Greek side pays more: on the full 5,156-item benchmark the language-matched checkpoint scores 73.7 against the base’s 80.6, a -6.9 pp raw-lane cost (this is the raw-lane reading; the best-of-15 arm reads -0.7 on the 1,000-item probe under answer-format compliance, §7). That figure sits inside the 7.7 pp seed floor measured on this family and decoding (§4), so we report it as a measured single-run cost and explicitly not as a property of the recipe. So the capability was never destroyed by teaching Greek; it was suppressed in the instruction channel (on the families where it returns at all, §10.1) by a recipe that targeted a single output language, while the question-following default survived on the recipe’s sibling arms (Table 12). Matching the language of the training pair to the language of the question keeps that default on every family, and reopens the instruction channel on two of the four (§10.1). Beyond the default, the explicit override (the capability whose loss Table 10 documents) also returns, partially: re-running the probe that produced 0/1,000 on the released language-matched checkpoints, an instruction to reason in English on a Greek question is now obeyed on 44.8% of items (Qwen; 62.5% on Gpt-OSS), with 24.8% still fully Greek and the rest mixed-script, and the reverse instruction (reason in Greek on an English question, $n=1,100$) yields Greek traces on 83.7% / 93.3% (the reverse lane of the same probe; Table 10 carries the Greek-question rows). Steerability is restored in kind, not in full, and, as §10.1 shows, not on every fam-

ily. Two readings we cannot exclude: partial compliance may be no more than the matched corpus putting English traces back in-support (a distribution shift rather than restored instruction-following), and the family ordering (62.5% over 44.8%) reproduces the families’ steerability gap under the one-directional recipe (95% vs 0%, §13), so part of the compliance is plausibly the family’s, not the recipe’s. The word the paper is allowed to use is therefore **language-matching**, not language-locking: the reasoning language follows the question.

10.1. The instruction channel re-opens on two families, not four

Running the identical probe, scorer and thresholds on the two Nemotron language-matched releases completes the picture, and it is not the one the two-family result suggested:

released checkpoint	“in English” [↑] (Greek Q)	“in Greek” [↑] (English Q)
Sophea-Qwen3.6-v1	44.8%	83.7%
Sophea-Qwen3.6-v1.1 (RLVR)	53.9%	85.7%
Sophea-OSS-v1	62.5%	93.3%
Sophea-Nemo-3.5-Lightning-v1	0.0%	92.7%
Sophea-Nemo-3-Nano-v1	0.0%	87.5%
<i>one-directional (reference)</i>	<i>0.0%</i>	<i>91.4%</i>

Table 11: Instructed-override compliance on the five released checkpoints: the share of traces in the instructed language when the instruction contradicts the question ($n=1,000$ Greek-question and $n=1,100$ English-question items; same scorer as Table 10). Every release obeys “switch to Greek”; only the Qwen line and Gpt-OSS obey “switch to English”, and the RLVR refinement widens Qwen’s opening (44.8 $\rightarrow$ 53.9%, §15.1). Two instrument notes, both checked: the Nemotron dumps do not separate trace from answer, so the mandatory Greek answer line falls inside the scored text: re-scoring with that line stripped leaves both Nemotron arms at 0.0% (the Qwen control moves 44.8 $\rightarrow$ 49.5%, the expected direction); and the Nemotron arms return 5–9% of rows too short to score against Qwen’s 0%, consistent with the low reasoning ceiling reported in §13.

So the instruction channel is *not* a property the language-matched recipe restores. It restores the question-following default on every family we trained (Table 7), and re-opens the override on two of four checkpoints. The asymmetry is uniform in one direction and family-dependent in the other: every release complies with an instruction to reason in Greek (83.7–93.3%), none complies fully with an instruction to reason in English, and the two Nemotron arms do not comply at all. The split tracks neither architecture (the two Nemotron generations differ from each other in reasoning ceiling but agree here) nor recipe (identical). What we can now

say (from the pre-registered RLVR round of §15) is that the channel *responds to a verifiable reward*: an override-obedience term moves compliance +9.1 pp on this family with a flat random-reward control and no cost on any held direction, though short of the pre-registered trainability bar (§15.1). Why the channel sits open on two families and shut on two others after identical SFT remains the open question; whether it moves under a reward at all no longer is.

The practical consequence is the part a deployment can act on. Trace-language *steerability* is not a property this recipe confers, and it is not predictable from the axes a model card publishes: the four checkpoints here share a corpus, a recipe, an adapter configuration and a serving budget, agree on every default-behaviour metric we report, and still split 62.5/44.8/0/0 on this one. **A deployment that needs the trace language to be steerable at inference (a bilingual support desk, a reviewer-facing audit mode, any product where the operator overrides the user’s language) must test this axis per checkpoint, and must not infer it from the recipe, the family, or the question-following behaviour that this paper otherwise shows generalising.** The probe costs one instructed pass over a benchmark that already exists ($n=1,000$ per direction here), which puts it in the same class as the seed control of §4: cheap, and it changes a claim we would otherwise have made. We add it to the supported column of Table 15 on those terms.

The honest reading of Table 10 is therefore not “the model forgot English” but “a model that targets a single output language learns to ignore explicit language instructions.” Note the base is partially locked too (told to reason in Greek it still produces English 72% of the time), so asymmetric locking is not unique to fine-tuning; what is unique is how firm ours became.

The finding is not Qwen-specific, and it is not specific to the language-matched checkpoint. Running the same 1,100-item English control (the GSM8K originals of the Greek math axis, anchored-extraction scoring, §8) on the release candidates of all three families, every fine-tuned arm still reasons in English on 100% of English questions (median Greek-character ratio 0.00), at +0.6 to +2.2 pp against base for the reasoning-recipe arms; the matched-recipe rows stand at +0.4 (Qwen) and −3.6 (Gpt-OSS) against their own bases, with the Gpt-OSS English retention since re-measured at +1.0 macro against base (Table 13):

A question the printed numbers force: on our printed axes the one-directional fine-tunes match or beat the matched checkpoint: higher Greek-trace fidelity (98.7 vs 97.98), the same 100% English-on-English default (theirs measured on sibling arms of the same recipe, Table 12; no single one-directional arm carries all three numbers

arm ($n=1,100$)	accuracy	reasons in EN
Qwen base / ft	94.0 / 96.2	100% / 100%
Gpt-OSS base / ft	95.5 / 96.1	100% / 100%
NemotronH base / ft	92.7 / 93.7	100% / 100%
Qwen language-matched	94.4	100%
Gpt-OSS language-matched	91.9	100%

Table 12: English questions get English traces from every arm measured. The fine-tunes in the first three rows are the dual-mode / reasoning-recipe checkpoints (§7), kept deliberately: they are the recipe-class evidence that one-directional training does not lock English *questions*. The last two rows are the language-matched checkpoints: the released Qwen checkpoint, re-run on this exact control (94.4, +0.4 over its base, 100% English traces, consistent with its 1,900-item probe above), and the Gpt-OSS language-matched arm evaluated after merging (the pre-repair arm; the released, format-repaired checkpoint’s control has not been generatively re-run, Table 7). Together the rows confirm the language-matching result is not Qwen-specific; none of this retests the hard one-directional lock of Table 10, which remains a property of training in a single output direction.

at once), and no -6.9 print. What they lack is the instruction channel: the override probe reads 0/1,000 on the one-directional arm and 44.8% English compliance on the matched checkpoint (62.5% on the Gpt-OSS release; Table 10). We release the matched checkpoints as the headline because default behaviour and instructability are different axes: the matched recipe is the only one measured to restore the default on every family, and the only one under which the override returns at all, though on two of four checkpoints, not four (Table 11). The release rationale therefore stands as “defaults kept everywhere, overrides re-opened on two families”: measured, no longer a hedge, and no longer stated as a property of the recipe alone.

11. Two pre-registered questions, and how they closed

The language-matching result of §10 answered the first of the three questions we pre-registered. The other two closed since, each by a cheap evaluation whose design was fixed before the result existed; we state each design first and its answer second, so the results read as a test rather than a story fitted afterwards.

E2: Did Greek reasoning SFT damage general ability, in either language? Answered: no on the earlier recipe, with one repairable exception on the language-matched recipe. (One prior frames the null: LoRA is documented to forget less than full fine-tuning at matched target gain [3], so flat macros are the expected case, not a surprise; reported anyway, because a 35B MoE at rank 32 on a translated suite is a differ-

ent setting from theirs.) The corpus carries 5,211 English rows precisely as replay, and the risk on the Greek side is subtler: a model taught to reason at length may lose the short, direct competence that general benchmarks measure. The instrument is fixed by the data, not chosen for convenience. Every arm’s non-reasoning half *is* the **Sophea-Titan-1** mix (§2), so the retention question has to be asked on the suite that model was released against, or the numbers compare to nothing: nine frozen Greek benchmarks (GreekMMLU, MMLU-el, HellaSwag-el, ARC-Easy/Challenge-el, Belebele-el, Winogrande-el, TruthfulQA-el, Medical-MCQA-el) and five English-retention benchmarks (MMLU, HellaSwag, ARC-Easy/Challenge, Winogrande), all scored by log-likelihood with no generation involved. Two macros, deliberately not averaged into one: the Greek macro answers whether reasoning SFT cost general Greek ability, the English macro whether it cost English.

Greek macro	base	language-matched	+ format repair
Gpt-OSS	61.2	53.9	58.0
<i>macro deltas vs base (points):</i>			
Sophea-Qwen3.6-v1		−0.01 Greek / +0.08 English	
Sophea-OSS-v1		−3.2 Greek / +1.0 English [†]	
Sophea-Nemo-3-Nano-v1		+3.8 Greek / −2.0 English	
Sophea-Nemo-3.5-Lightning-v1		+1.7 Greek / −1.1 English	

Table 13: Forgetting on the Titan-1 suite. Top: Gpt-OSS on the nine Greek non-reasoning benchmarks, base against the language-matched fine-tune, before and after the format-repair dose: the language-matched fine-tune initially costs -7.3 points, and the repair dose recovers 4.1 of them, so the residual is -3.2 not zero. Below: the macro deltas this paper reports per family, fine-tuned arm vs its own base; **the format-repaired Gpt-OSS is the fine-tuned arm we stand behind.** Qwen is flat in both languages; Nano *gains* Greek (+3.8) against the family’s low 0.49 Nano base (mostly catch-up, not forgetting); Lightning is read against its *own* base, measured separately (Greek macro 57.5, well above Nano’s 48.6), and gains +1.7; an earlier +10.6 reading against the shared Nano base is superseded by that measurement. Per benchmark against binomial standard error; a macro delta is a direction, not a test. [†]On English the pre-repair language-matched Gpt-OSS lost -7.7 points ($75.7 \rightarrow 68.0$); re-measured on the repaired release, the English macro is 76.7, +1.0 *above* base: the repair dose recovered both lanes, so the pre-repair loss was format behaviour throughout. Macro deltas are computed on unrounded macros; Table 14 prints one decimal, so its Qwen columns round to $75.7 \rightarrow 75.6$ Greek / $85.1 \rightarrow 85.2$ English.

Table 13 reports the release reasoning models this paper stands behind (Table 14 opens every macro into its per-benchmark readings; Figure 10 draws the same deltas as one map). A reader comparing the *base* columns against the numbers the labs publish for the same checkpoints should expect small offsets, on the order of $\pm 1-2$ points: published tables come from different evaluation

benchmark	Qwen		Gpt-OSS		Nemotron			
	base	ft	base	ft	Nano base	Nano ft	Ltng. base	Ltng. ft
<i>Greek (nine benchmarks, the Greek macro of Table 13):</i>								
ARC-Challenge-el	91.8	92.0	77.1	68.6	54.2	59.0	68.7	72.8
ARC-Easy-el	96.9	97.1	85.4	79.9	62.3	70.1	75.8	81.4
Belebele-el	92.7	93.7	86.0	81.3	67.0	78.0	81.3	83.3
GreekMMLU	83.8	83.0	66.3	61.3	58.8	53.3	64.2	62.2
HellaSwag-el	57.6	59.4	42.9	52.1	39.2	47.6	44.5	51.2
Medical-MCQA-el	81.2	80.3	42.1	38.7	27.8	31.2	41.7	38.2
TruthfulQA-el	39.1	36.5	39.7	35.5	31.7	33.2	34.2	33.7
Winogrande-el	61.0	62.5	55.4	59.6	53.3	58.3	55.8	60.1
MMLU-el	76.9	76.2	55.5	45.4	43.2	40.8	51.3	49.6
<i>Greek macro</i>	<i>75.7</i>	<i>75.6</i>	<i>61.2</i>	<i>58.0</i>	<i>48.6</i>	<i>52.4</i>	<i>57.5</i>	<i>59.2</i>
<i>English retention (five benchmarks):</i>								
ARC-Challenge	96.1	95.7	90.0	86.8	88.3	82.2	92.0	89.1
ARC-Easy	99.1	99.1	96.0	95.7	96.2	93.4	97.8	96.4
HellaSwag	73.2	74.2	57.0	68.4	67.9	73.1	72.7	75.9
Winogrande	73.2	74.2	64.6	66.7	69.9	72.9	72.0	73.3
MMLU	83.9	82.6	70.8	65.8	69.9	60.5	75.8	69.7
<i>English macro</i>	<i>85.1</i>	<i>85.2</i>	<i>75.7</i>	<i>76.7</i>	<i>78.4</i>	<i>76.4</i>	<i>82.0</i>	<i>80.9</i>

Table 14: Every benchmark behind the macros of Table 13 (Titan-1 suite, non-reasoning mode; accuracy %). Qwen *ft* is Sophea-Qwen3.6-v1 (language-matched); Gpt-OSS *ft* is Sophea-OSS-v1, the format-repaired release (its pre-repair intermediate is in Table 13); Nano *ft* is Sophea-Nemo-3-Nano-v1, the language-matched Nano release; Lightning *ft* is Sophea-Nemo-3.5-Lightning-v1 (language-matched); the two Nemotron generations share the same 31.6B / 6-of-128 routing profile (Table 1) and each is read against its own base column. (The earlier one-directional recipe arm is reported only in §13’s recipe history, not as a release.) The per-benchmark view localises the Gpt-OSS story: before the repair the loss concentrated on the instruction-format-heavy reading benchmarks (Belebele-el -17.7 , ARC-Challenge -13.1 el / -19.1 en, MMLU-en -16.0) while HellaSwag and Winogrande *gained* in both languages, and the repair dose recovers most where the loss was format (Belebele-el regains $+13.0$ of the 17.7). English retention on the released Gpt-OSS checkpoint recovers to $+1.0$ above its base; Lightning retains English within 1.1 points of its own base.

stacks: serving engine, prompt template, few-shot count, sampling parameters (temperature, top- p) where generation is involved, and answer-extraction rules. None of those differences is a property of the model. Every arm in these tables is scored by the same harness under the same settings, so the columns are internally comparable; it is the base-to-fine-tune *delta*, not the absolute level, that this section’s claims rest on. Sophea-OSS-v1 is the column that matters. Sophea-Qwen3.6-v1 is flat to a tenth of a point in both languages. On Sophea-OSS-v1 the language-matched fine-tune is not uniformly free: it costs -7.3 points Greek and -7.7 English (macro $75.7 \rightarrow 68.0$) *in both reasoning and direct modes*, with a 70% answer-fallback rate. Adding a format-repair dose (the identical mix plus a 2% dose of trace-less anchored answer-closing twins) recovers 4.1 of the Greek points and collapses fallback from 70% to 26% on the full benchmark (per-domain in Table 6), so the Sophea-OSS-v1 deficit is format-closing behaviour, fixable by data, not a capability the fine-tune destroys. We therefore release **the format-repaired Gpt-OSS as Sophea-OSS-v1**; the cost that survives is -3.2 Greek. English retention on the released checkpoint, re-measured on the same five benchmarks, reads 76.7 ($+1.0$ above its base), so the pre-

repair -7.7 was the same format-closing behaviour on the English lane, and the Greek-side repair dose recovered both. The replay did its job on Sophea-Qwen3.6-v1 and Sophea-Nemo-3-Nano-v1, and the language lock in E1 is a preference, not damage; the single family-specific cost is real but repairable.

Register and grammar, probed directly. Two small judge-scored probes ask what the NLU macros cannot: does the fine-tune still switch address register on request (the formal plural-of-politeness vs the informal singular; 46 items spanning single-turn, rewrite and mid-conversation switches), and is its free-form Greek grammatical (gender/number/case agreement, aspect, voice/mood, clitic placement; 58 items)? Register control survives fine-tuning in every family and improves where the base was weakest (the two Nemotron bases go 40/46 and 37/46 to 44/45 and 43/46; Qwen dips $45 \rightarrow 42$, Gpt-OSS is flat at 39/46). Grammaticality roughly *doubles* on both Nemotron fine-tunes ($13/58 \rightarrow 27$ and $12/58 \rightarrow 29$, driven by the agreement axis), is flat on Qwen ($41 \rightarrow 42$), and improves on Gpt-OSS ($32 \rightarrow 40$). The Gpt-OSS cell also demonstrates why these probes need adversarial reading: its first-pass score was 23/58, a spurious regression caused by the same channel-marker

Retention: fine-tuned minus its own base (pp)
green = gained, plum = lost

ARC-Challenge-el	+0.2	-8.5	+4.8	+4.1	Greek (9)
ARC-Easy-el	+0.2	-5.5	+7.8	+5.6	
Belebele-el	+1.0	-4.7	+11.0	+2.0	
GreekMMLU	-0.8	-5.0	-5.5	-2.0	
HellaSwag-el	+1.8	+9.2	+8.4	+6.7	
Medical-MCQA-el	-0.9	-3.4	+3.4	-3.5	
TruthfulQA-el	-2.6	-4.2	+1.5	-0.5	
Winogrande-el	+1.5	+4.2	+5.0	+4.3	
MMLU-el	-0.7	-10.1	-2.4	-1.7	
ARC-Challenge	-0.4	-3.2	-6.1	-2.9	English (5)
ARC-Easy	+0.0	-0.3	-2.8	-1.4	
HellaSwag	+1.0	+11.4	+5.2	+3.2	
Winogrande	+1.0	+2.1	+3.0	+1.3	
MMLU	-1.3	-5.0	-9.4	-6.1	
	Sophea-Qwen-v1	Sophea-OSS-v1	Sophea-Nemo-Nano-v1	Sophea-Nemo-Lightning-v1	

Figure 10: Every per-benchmark retention delta behind Table 14: fine-tuned release minus *its own* base (Lightning against the Lightning base; Sophea-OSS-v1 is the format-repaired release), Titan-1 suite, non-reasoning mode. Per-benchmark binomial sigma applies and a delta is a direction, not a test (Table 13); the value is the pattern: the Qwen column is near-white in both languages, the Sophea-OSS-v1 residual concentrates on the knowledge-heavy benchmarks (MMLU-el -10.1) while HellaSwag gains in both languages on every family, and MMLU is the largest English loss for all three non-Qwen releases.

behaviour documented above: the fine-tune prefixes its text with a bare channel marker, the judge read the marker as ungrammatical Greek, and stripping it flipped the verdict from “worst fine-tune” to “above base”. At $n=46/58$ these are counts, not rates, and the scorer is itself an LLM judge (temperature 0), so we report them as directional evidence only: the behavioural shift this paper documents costs neither politeness control nor grammatical Greek in any family.

E3: Does the non-reasoning mode still work?

Answered: yes; dropping the direct half costs nothing measurable there. The direct half exists to preserve that mode, its removal is our largest measured accuracy effect, and every other number in this paper comes from reasoning benchmarks, so the design

was fixed in advance: run the Greek NLU benchmark (instrument (v), §2) in direct mode, no reasoning requested, across BASE, REASONING (which never saw the direct half) and TWO-PHASE (which did). This was the one place where dropping phase 2 should *cost* something. It does not: on strict scoring Qwen reads 0.680 base, 0.683 reasoning-only, 0.696 two-phase (a gap inside the seed floor), and on Gpt-OSS the reasoning-only arm sits at base level (-1.6 pp) while *two-phase* is the only arm that degrades (-9.0 pp strict, and the one format break: its direct-mode outputs carry English reasoning prose the strict parser rejects). The release checkpoints were then re-run on the Titan-1 suite in the same mode: Qwen flat (-0.1), Nemotron-3.5 $+1.7$ against its own base (an interim $+10.6$ reading, taken against the shared Nano base before the Lightning base was measured, is superseded), and pre-repair Gpt-OSS -7.3 : the same -7.3 as its think mode, so that family’s deficit is mode-independent, exactly what the format-repair reading of Table 13 predicts. Two by-products belong in the record. The `<think>`-carrier families have a clean off switch: zero unrequested traces in 9,751 direct-mode generations. The Gpt-OSS *base* has no off switch at all (it opens its analysis channel on 100% of rows even with reasoning effort set to none), and neither of its fine-tunes installed a switch the base never had.

Why these designs. Each is decidable by evaluation alone, needs no retraining, and would change a sentence this paper commits to. (E1, the question that recast the goal from locking to matching, is answered in §10.) E2 closed the forgetting question: general ability was flat or gained on Sophea-Qwen3.6-v1 and Sophea-Nemo-3-Nano-v1, and the one release that lost ground (Sophea-OSS-v1) was shown to lose it to format-closing behaviour and to recover under a format-repair dose, not to a destroyed capability. E3 closed the direct-mode question the same way: the design was fixed while either outcome was still possible, and the answer (dropping the direct half costs nothing measurable on the non-reasoning mode) landed on the side that strengthens §14 rather than the side we could not have walked back.

12. Per domain: indistinguishable on accuracy, not on cost

Figure 11 is deliberately not a leaderboard. At a 7.7 pp seed range, naming a per-domain winner selects the top of a noisy draw, and a rerun would name someone else; the shaded band makes that visible rather than leaving it to a footnote. Read that way it says two things.

On accuracy, no per-domain winner is readable against the floor. Math 87.6–95.6, commonsense 70.7–82.3, logic 40.5–46.3: spreads of 8.0, 11.6 and 5.8 pp against a 7.7 pp seed range, two of them slightly wider

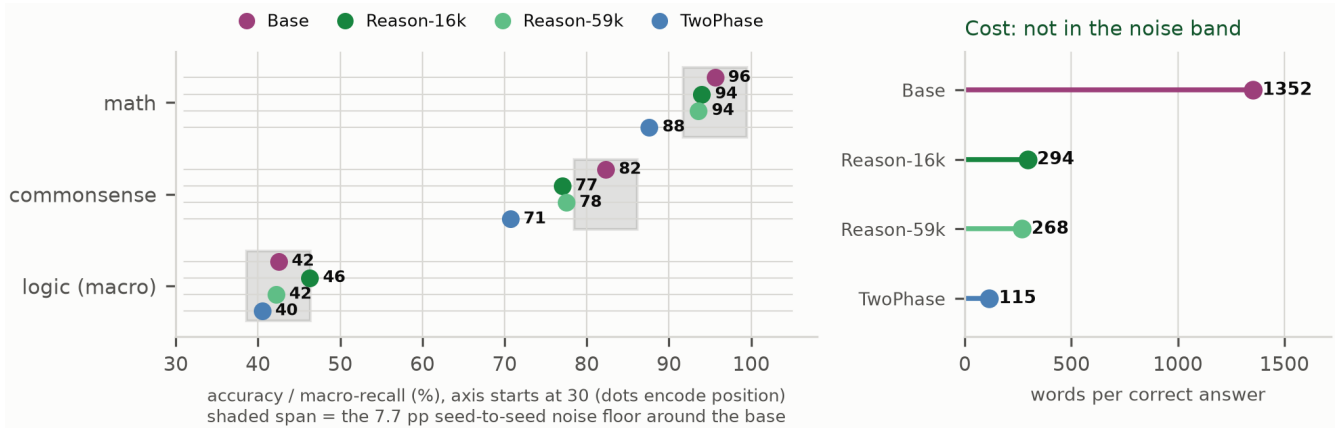


Figure 11: Left: accuracy per axis on the 5,156-item benchmark for the *Qwen recipe arms* (the cross-family releases are Table 6), with the seed-to-seed noise floor shaded around the Qwen base. Right: the same arms’ cost in *words* per correct answer (token-denominated cost is Figure 5). Math uses the 250 human-translated items only (§8); logic is macro-recall (§3).

than the range itself. Choosing a model per domain on these numbers is not supported by this evidence.

On cost, the arms are not close. 1,352 *words* per correct answer for the base against 268–294 for the reasoning-only fine-tunes: a 4.6–5.0 $\times$ difference, far outside anything seeds do, though in tokenizer tokens the gap narrows or reverses by family (§5). If a deployment must pick one Greek reasoning model, the defensible basis is cost and trace language, not per-axis accuracy.

13. What each family taught us

The three models behaved differently in ways that no accuracy table records, and that anyone adapting them will meet. We report these because they cost us real time and are absent from the model cards. Library versions are peft 0.19.1 / transformers 5.12.1 throughout; items marked *[env]* are version-specific rather than properties of the model.

13.1. Qwen3.6-35B-A3B: fused experts, and a base that hides behind its token budget

[env] Its experts are **stacked parameters**, so LoRA attaches via `target_parameters` (28 tensors). That adapter **cannot be loaded back** by `PeftModel.from_pretrained`: the call raises inside a weight converter, and the only route to evaluation is merging the adapter into dense weights first. We lost an entire evaluation round to this before noticing that the base arms had run and the fine-tuned arms had not.

Its base is truncation-crippled in think mode. At 1,536 tokens, 670 of 1,000 rows never reach a final answer and it scores 20.8; at 4,096 it scores 77.2 (the 1,000-item probe’s three-axis mean, as in §1), above its own direct mode. Any comparison against this base at a standard budget measures the budget. Its traces run 4,000–

5,400 characters where the fine-tune’s run 150 words.

It is the most seed-sensitive of the three (sd 4.4 pp, range 7.7), and its *one-directional* fine-tune is completely **language-locked**: told to reason in English it produces Greek on 1000/1000 items (the released language-matched checkpoint complies on 44.8%, Table 11; 53.9% after the RLVR round, Table 16). The base is partially locked the other way: told to reason in Greek it still emits English 72% of the time.

13.2. Gpt-OSS-20B: a different trace carrier, and it never stops reasoning

It does not use <think>. Reasoning arrives in *harmony channels* (*analysis / assistantfinal*), so a scorer written for one carrier silently mis-reads the other.

It emits an analysis channel even when reasoning is switched off: 1000/1000 rows at `reasoning_effort=None`. We gated the channel split on having *asked* for reasoning rather than on the markers being present, so every direct-mode number was computed over trace and answer concatenated, and 95 logic rows scored as unparseable. Split whenever the markers appear.

[env] A **detailed thinking off** system message does not disable reasoning; it returns `content=null` with the text in *reasoning*. Use `chat_template_kwargs`. It also rejects `sdpa` and needs **eager** attention.

It is the counterexample twice over. Dropping the non-reasoning half of the corpus helps Qwen by +6.9 pp and *hurts* Gpt-OSS by 5.6 (62.5 vs 68.1, one run per condition, §7); and its fine-tune stays **steerable**, complying with an English-reasoning instruction 95% of the time where Qwen’s complies 0% (the released language-matched arms narrow the gap but keep the ordering: 62.5% vs 44.8%, §10). Same recipe, same corpus, opposite behaviour. We have no mechanism for either dif-

ference, though §15.1 shows the override quantity itself responds to a verifiable reward.

13.3. NemotronH-30B-A3B: a hybrid that fights every default

Its experts are individually materialised as `nn.Linear`, so PEFT attaches one adapter *per expert*: **4,188 adapter tensors** against 28 for Qwen, at *fewer* trainable parameters (325M vs 688M). Step time then goes on kernel launches and gradient all-reduces rather than FLOPs; it trains $3.7\text{--}7.6\times$ slower than the others. Reducing adapter count $7\times$ bought only 23%, so this is not the whole story; we did not isolate the rest.

[env] Three defaults have to change, none of which announces itself. `ddp_find_unused_parameters` must be **True** (routing leaves experts without gradient, and DDP aborts mid-run); `model_accepts_loss_kwargs` must be **False** (NemotronH ignores `num_items_in_batch`, so loss and gradients otherwise scale with the accumulation count); and evaluation needs `trust_remote_code=False`, because the shipped modeling file *trains* correctly but its `prepare_inputs_for_generation` indexes a `cache_position` that `generate()` passes as **None**. That last one fails only at inference, so five hours of training completed before it surfaced.

[env] `mamba-ssm` must be installed or 23 of its 52 layers fall back to a pure-PyTorch scan; transformers logs this on every run. Installing it changed step time by 0%; worth recording, because the obvious diagnosis was wrong.

And the claim replicates on it. This is the third, non-transformer architecture, and the headline result holds: its base reasons in Greek on 0% of the Greek benchmark (median ratio 0.23), its fine-tune on 98%, at an accuracy cost of -0.6 pp, with the generation-cap truncation rate falling $10.8 \rightarrow 2.6\%$ and measure loops in its traces dropping from 15.0 to 0.0 per thousand rows (same 5,156-item benchmark, full lanes on both arms). Whatever made the reasoning language movable on two transformer MoEs is not a transformer property; on this evidence it travels with a strong base plus a language-directed corpus.

The pattern. Every family needed a different fix, none of the fixes was announced by an error at the point of the mistake, and three of them produced *plausible numbers* rather than crashes. A pipeline that runs to completion on a new model family is not evidence that it ran correctly.

14. What we can and cannot recommend

Table 15 states every configuration recommendation this evidence supports, and the ones it cannot; the basis col-

umn is the point, because a configuration table without it would be the overclaim this paper argues against.

15. A third pre-registration: verifiable-reward RL, designed before its numbers

Everything this paper measures ends at the boundary of imitation. SFT reproduces the traces it is shown, and three of our measured defects are structural to that: the released Qwen checkpoint fails to emit the requested answer line on 24% of Greek items and leaks the answer into the trace channel on 3.5% (§3), and clean demonstration plausibly cannot push a leak *rate* to zero, because the imitation objective never sees the counterfactual. The third is the one §10.1 leaves without a mechanism: after language-matched training the explicit override instruction is obeyed on 44.8% of items on this family, and we do not know what makes obedience trainable. All three quantities are deterministically checkable, which makes them exactly the setting for reinforcement learning with verifiable rewards (RLVR) [9, 29]: the objective can be written without a learned reward model or an LLM judge, so the reward cannot drift, cannot be sycophantic, and can be audited line by line. This section pre-registers that experiment the way §11 pre-registered the paper’s two questions: design, reward, defenses and decision rules are frozen here, and were frozen (rules, reward, fuzz suite and arms) *before* the first optimizer step. The outcomes, measured after that freeze, close the section (§15.1).

The risk the design must carry is also on record: GRPO-style optimization is a known amplifier of cross-lingual drift, with reasoning reverting to the model’s dominant language as training progresses [22]; at small dense scale, an accuracy-only reward collapsed language consistency to zero, and a language-consistency term repaired it at no accuracy cost [42]. Whether that repair holds when the property being protected was installed by SFT, on a mixture-of-experts model, through a LoRA adapter, is untested, and our headline result (§10) is precisely what an accuracy-only gradient would erode first.

Objective and environment. GRPO [29] on the released Qwen checkpoint: 8 sampled completions per prompt, group-relative advantage, no KL term ($\beta=0$, following 39), constant learning rate 10^{-6} , temperature 0.7, completion budget 1,536 tokens. The budget is a measured decision, not a default: at 1,024 tokens roughly 90% of completions truncated, and because a truncated completion is gated to zero correctness (below), the correctness term was starved and the run optimized termination and format only. Adaptation is LoRA [16] at $r=32$, $\alpha=64$ on the fused expert projections, the same fused-parameter targeting the SFT used (§13), with dropout fixed at 0: under a policy-gradient objective, dropout

Supported by our evidence	basis
Drop the hybrid phase 2; train reasoning-only	+6.9 pp over 15 runs, $p = 0.0008$ (§7); and the direct mode it sacrifices costs nothing measurable on direct-mode NLU (E3, §11)
Generate exploratory traces, not explanations	structure 0.27 $\rightarrow$ 0.68 on the same questions
Forbid formulaic openings in the generator	24/24 identical openings without it
Report fallback% beside accuracy	separates reasoning from format compliance
Run the seed control before any ablation	7.7 pp (§4)
Probe trace-language steerability per checkpoint	same recipe, four checkpoints, 62.5/44.8/0/0 (§10.1)
Keep a human-translated benchmark subset	caught a 3.1σ artifact (§8)
Test numeric normalisation in the target language’s own convention	5 of 6 Greek-thousands answers scored wrong while right (§8.5)
Gate metrics on length correlation	removed 3 of our 7
Fix format and leak defects with verifiable-reward RL, not more SFT data	fallback 24 $\rightarrow$ 2.5%, leak 3.5 $\rightarrow$ 0%, control flat, $\sim$ 78 GPU-h/arm (§15.1)
Run a random-reward control arm with any RLVR claim	it reproduced our baseline on every axis; without it, every gain here would be unattributable (§15.1)
NOT supported	why
A specific corpus version	five expansions, all inside the noise floor
A data-selection method	selection $\approx$ random, -0.5 pp (0.31σ)
A LoRA rank, stride, or epoch count	never varied; no evidence either way
A per-domain model choice	two of three per-domain score spreads exceed the seed range itself (§12)

Table 15: Supported and unsupported configuration choices, each with its evidential basis.

makes the update policy differ stochastically from the rollout policy and corrupts the importance ratio, so a regularizer that is habit in SFT is a correctness bug here. All reward arms run simultaneously on identical hardware with the same data order and the same seed, so the only factor that varies between arms is the reward.

The prompt pool. Problems enter in language-matched *pairs*, the same item once in Greek and once in English, exactly as the SFT corpus was built (§10), because a Greek-only pool with any language-shaped term aims a forgetting gradient at the paper’s central property. A held-out slice carries the explicit override instruction, with its phrasing copied verbatim from the evaluation probe of §10.1: both contradicting directions (Greek question instructing English, and the reverse) *and* both agreeing controls, so the learnable target is the instruction, not the flip. Gold answers are numeric only, verified by the locale-aware parser below; the pool is deduplicated and decontaminated by 13-gram overlap against both the SFT corpus and every evaluation benchmark, in both languages (§8 is why in both).

A reward with no judge. Five deterministic terms: (i) *correctness*, the anchored answer equals gold under a locale-aware numeric parser (in Greek convention 17.500 is seventeen and a half thousand and 3.5 is three and a half, and §8.5 documents a scorer of ours that misread exactly this); (ii) *language consistency*, trace matches

the question’s language (redirected to the *instructed* language whenever an explicit instruction is present, so the two terms cannot both be satisfied by ignoring the instruction); (iii) *format*, the requested answer line is present; (iv) *termination*, generation ends inside the budget (a truncated completion scores zero correctness, because a right number inside an unfinished trace is not a solved problem); and (v) *override obedience* on the instructed slice. Two gates run underneath: language terms require a minimum of letters after stripping code and $\text{\LaTeX}$ (§8.5); and every behavioural term requires the trace to contain *work*, an intermediate value that is not the answer restated, deliberately not a length test.

Adversarial pre-flight, because a found exploit is permanent. A policy that discovers a reward bug encodes it in weights; unlike an analysis bug, it cannot be fixed afterwards. So the reward ships with an adversarial suite of 28 attacks that must pass before any rollout: trace elision (answer line only), fluent Greek filler with no computation and its $10\times$ -length variant, answer shotguns and number sprays against the extractor, code-block laundering of English content, the locale exploit above, truncation mid-number, disobeying and obeying override pairs (the disobeying trace must score strictly lower), and a robustness family that must not crash. Building this suite caught three exploits in our own draft reward before training (a fluent no-computation trace at

full marks among them) and motivated the work gate. One failure mode the suite structurally cannot catch is a term whose *input* never arrives: a reward called without its instruction field silently scores the term at zero and the arm degenerates into a different arm with no error raised; we caught exactly this in a pilot, and the trainer now refuses to start if the override arm’s pool carries no instructed rows, and asserts after training that instructed rows actually reached the reward. During training we store every rollout (prompt, all 8 completions, and the per-term score breakdown), so any suspicious number can be traced to the text that earned it.

Design: one varied factor, and a control that can veto everything. Four arms, identical in data, steps, seed and hardware, differing only in active reward terms: correctness+format +termination (the collapse probe: what does an accuracy-shaped gradient cost the language property?); the same plus language consistency (the candidate recipe); the same plus override obedience on the instructed slice (the mechanism probe for §10.1); and a *random-reward control*: same steps, same data, reward drawn uniformly at random. The control is not decoration: random rewards have been shown to recover most of an RLVR gain on this model family [28], so any axis on which the control matches a trained arm is elicitation, not learning, and the pre-registered rule is that the result on that axis is withdrawn regardless of what the trained arms show.

Decision rules, frozen before the numbers. Fidelity is a *gate*, not a tradable term: no outcome that buys a metric with Greek-trace fidelity below 97.0% counts as a win. Format fallback must reach $\leq 10\%$ and answer-channel leak $\leq 1.5\%$ to claim that RLVR closes defects SFT did not (24% and 3.5% baselines); the override question carries a three-branch rule: a rise below +5 pp is the null, a rise of ≥ 15 pp reaching $\geq 60\%$ from 44.8%, with the reverse direction held at $\geq 80\%$, is the only outcome that may be called *trainable*, and anything between is reported as reward-responsive with the missed bar stated (the full frozen rule table is archived with the released artifacts); accuracy movements inside the ± 7.7 pp seed floor of §4 are reported as inside the floor, never as gains; and every claim is measured on the held-out instruments of §3, never on the training reward, which is the quantity a reward hack inflates. The null branches are pre-committed too: if the floors do not drop, the defects are reported as not reward-addressable at this budget; if override does not move, the finding of §10.1 keeps its “no mechanism” framing, strengthened. Abort conditions (entropy collapse, response length exceeding twice baseline, reward rising with fallback, any term saturating early) stop a run before it can manufacture a result.

15.1. The outcomes, against the frozen rules

The round ran as designed: 2,000 prompts (800 per language, paired, plus a 400-row override slice), one epoch, 8 completions per prompt, four arms in parallel on identical hardware with the same seed and data order, ~ 39 hours. No abort condition fired; the fuzz suite passed before launch; the override slice’s arrival at the reward function was asserted at run end on every arm. Table 16 holds every number a decision rule reads.

The floors drop, and the control proves it is learning. Fallback $24.1\% \rightarrow 2.5\%$ and answer-channel leak $3.53\% \rightarrow 0.00\%$ on the plain-RLVR arm, both past their pre-registered thresholds ($\leq 10\%$, $\leq 1.5\%$), with fidelity held. The random-reward control (same steps, same data) moves on neither (22.1%, 3.61%). The branch fires as written in advance: *in this round, RLVR closed the format and leak defects that our SFT did not*, and for the leak, plausibly could not (the counterfactual argument above), at ~ 78 GPU-hours per arm (39 hours wall on two GPUs) over a 2,000-prompt pool. On format, SFT is not helpless (a targeted repair dose cut another family’s fallback $70 \rightarrow 26\%$, §11), but it plateaued at 24% on this one, where the reward reaches 2.5%.

Steerability is reward-responsive; the trainability bar was not met. The override arm moves instructed compliance $44.8 \rightarrow 53.9\%$ (+9.1 pp; the control sits at 44.1, so the control-adjusted effect is +9.8 pp at $n=1,000$ per cell) while *holding* every hold-gate: reverse-direction compliance rises to 85.7% (rule: ≥ 80), the agreeing direction is untouched (98.5%, unchanged from baseline; same probe as Table 11, not shown in Table 16), fidelity is the highest of any arm (98.27). Under the frozen three-branch rule this is the middle branch, and we word it accordingly: the channel *responds* to a verifiable reward (the first mechanism-bearing movement of this quantity), but the $\geq +15$ pp / 60% trainability bar was missed, so *trainable* is a claim this round does not earn. One ordering we report without a mechanism: the +language arm gains least (45.9 vs 48.8 without the language term), consistent with a language-consistency pressure on the bulk of the pool opposing instructed switches, even though the reward redirects that term on instructed rows by construction.

The collapse did not come. The arm with no language protection ended at 98.22% Greek-trace fidelity, above its starting point, with zero in-question switches. At this scale and budget (a MoE adapted by LoRA, 16,000 completions, the property installed by SFT rather than by RL), the accuracy-shaped gradient did not erode language matching. We state this as bounded: it contradicts the dense-scale prior [22, 42] at our operating point, not everywhere.

arm	Greek		fall-		override		reverse	
	fid. %↑	acc↑	back %↓	leak %↓	el→EN %↑		en→EL %↑	
SFT checkpoint (before)	97.98	73.7	24.1	3.53	44.8		83.7	
random reward (control)	98.06	74.0	22.1	3.61	44.1		83.8	
correct+format+term.	98.22	77.2	2.5	0.00	48.8		84.7	
+ language	98.02	77.2	5.6	0.04	45.9		84.5	
+ override	98.27	77.0	2.8	0.02	53.9		85.7	

Table 16: The pre-registered RLVR round, scored on the held-out instruments (Greek think lane $n=5,156$; override lanes $n=1,000/1,100$), same scorer, same day. The *before* row is the released checkpoint’s frozen dump rescored with the current scorer; its full same-day regeneration reproduced all 5,156 responses *bit-identically* (greedy decoding is deterministic here), and the plain-mode lanes to the decimal (44.8/83.7): the before column is not a provenance caveat but a verified constant. The control row is the entire admissibility argument: on every axis the random-reward arm reproduces the baseline, so nothing below can be elicitation. Accuracy is shown for completeness only: +3.3–3.5 is inside the ± 7.7 seed floor of §4 and is claimed by nobody.

Finding 5. The pre-registered RLVR round moved what SFT did not, and the control shows it is learning: answer-format fallback 24.1 $\rightarrow$ 2.5% and answer-channel leak 3.53 $\rightarrow$ 0.00%, both past their frozen thresholds with fidelity held, while the random-reward arm, trained on the same steps and data, reproduced the baseline on every axis. The override channel responded (+9.1 pp, every hold-gate kept) but missed the frozen trainability bar; the language property survived an unprotected accuracy gradient.

Disclosures. The gates lanes were re-run mid-evaluation after the same-day baseline exposed a render flag silently disabling the reasoning trace: the sixth entry of §8, and the reason the *before* column’s provenance is stated in the caption. The override slice lives in the shared pool, so the +language $\rightarrow$ +override contrast is a dose comparison on one term, not presence/absence. And the scope is one family (Qwen), one seed per arm, the final checkpoint, an unscreened pool: these are existence proofs with a control, not recipes. The override arm passed every hold-gate (fidelity, both held directions) while missing the 60% override target, and ships as **Sopheia-Qwen3.6-v1.1**, alongside the SFT release it refines.

16. What to take away

For practitioners. Run the seed control before the ablations. One extra training run told us more than five corpus versions. Report **fallback%** or an equivalent compliance rate alongside accuracy; ours separated “cannot reason” from “will not answer in the requested form” and supplied the mechanism for our only surviving effect. Keep a human-translated subset of any machine-translated benchmark.

For the low-resource setting specifically. A fine-tune that leaves accuracy unchanged is not a failed fine-tune. Ours made a model reason in Greek instead of

English, with budget adaptation the base does not have, and, on the family whose traces shorten enough to repay Greek’s token fertility, at $3\times$ fewer tokens; on the others the trace-language gain costs token parity or a $1.6\times$ premium (§5). Those are the properties a deployment cares about, and an accuracy table cannot see any of them.

What we do not claim. No fine-tuned arm beats its base on the pooled Greek *reasoning* benchmark in any family (the NLU retention suite is a different quantity, where two arms gain, Table 13), and we report that as a measured null with a stated noise floor rather than an unresolved comparison. We report the fine-tunes’ behavioural advantages, and we report that the corpus work which consumed most of the project produced no measurable accuracy effect at all.

17. Related work

Reasoning language. Which language a multilingual model *should reason in* has an established literature, and the default answer is English. Shi et al. [31] show that English-language chain-of-thought outperforms native-language chain-of-thought on multilingual math, and Etxaniz et al. [11] report the same direction under self-translation: translate to English, reason, translate back. Wendler et al. [37] give it a mechanism: multilingual transformers’ latent space works in an English-like intermediate representation. Prompting interventions ride this current rather than oppose it: cross-lingual-thought prompting [17] machine-translates the problem into English and reasons there, and the translation-distillation line extends English CoT supervision to non-English tasks at scale [5, 24]. The closest point on the map is DeepSeek-R1 [9], whose RL phase adds a language-consistency reward precisely because unconstrained RL drifted into language mixing; the reward keeps the trace in the prompt’s language at a small measured cost to reasoning performance, treating a prompt-language trace as the success case rather than the failure, as does the consistency-reward work that followed [22, 42]. The

prompting and distillation work above makes the opposite call: reasoning in English is the success case and drift back to the prompt’s language the failure. Our deployment premise sides with R1’s trade and goes further: for a low-resource-language deployment of a reasoning model (here Greek) the trace *is* the product, and an opaque English trace is the defect, not the target. What we add over R1’s reward is the setting and the measurement: the property is installed by SFT alone in a mid-resource language, and we measure both directions of the resulting steering asymmetry (the locked fine-tune obeys “reason in English” on 0% of items; the base ignores “reason in Greek” on 72%, §10); that combination is where this paper sits.

Seed variance and reporting. The call to report variance rather than a best run is at least as old as the benchmark culture it critiques: Dodge et al. [10] argue expected-validation- performance budgeting, Bouthillier et al. [4] show baseline reorderings disappear once all sources of training randomness are marginalised, and Madaan et al. [20] quantify how evaluation choices (prompt, seed, few-shot draw) reorder LLM leaderboards. Our contribution to that line is not the call but two measured transfers: the effect size at *MoE + LoRA + a mid-resource language* scale (7.7 pp on the identical configuration, sd 4.4 pp, §4), and the asymmetry that the seed variance lands almost entirely on accuracy while our behavioural metrics sit flat across the same three seeds. The second transfer is what lets a paper this size survive its own noise floor: it is also the thing we have not seen measured in the seed-variance literature, which works almost entirely in accuracy or expert-human preference space.

Trace structure, selection, and the self-taught lineage. Our corpus-generation pipeline is a direct descendant of STaR [40]: sample a trace, keep it only if the final answer agrees with gold, discard the rest rather than repair them. What the trace itself should *look like* is laid out by Li et al. [19] (“the structure of Long CoT is critical to the learning process, whereas the content of individual reasoning steps has minimal impact”) and Gandhi et al. [13], whose reasoning-behaviour count (verification, backtracking, subgoal setting, backward chaining) names the variable our structure score was groping for. The cost side of traces has its own literature: the over-thinking phenomenon in long-CoT models [6] is what our budget-overflow metric (§5) quantifies per item rather than per benchmark. Our data-selection null is anticipated by Xia et al. [38], who found that self-scoring selection methods “struggled to significantly outperform random selection” at scale: the control we should have run first, and did run last.

Tokenizer fertility and the word-vs-token trap. That low-resource languages pay a tokenizer tax is a stan-

dard observation in multilingual evaluation; Wang et al. [36] quantify fertility gaps of the order we measure (2.3–2.5× for Greek against English on all three families, §5). The figure matters here because it is the hidden hand behind every cost comparison this paper makes: a word-denominated reasoning budget is the language-fair metric, and a token-denominated one silently double-counts the fertility tax (M3, §3; we report both). The same trap, units quietly re-priced per language, recurs whenever a derived quantity (cost per correct answer, budget overrun) inherits the units of a primary one.

The Greek line, and our instruments’ provenance. Greek NLP has moved past the era of translating nothing and measuring nothing: Meltemi [35] and Krikri [26] are the dedicated open Greek base models, and their evaluation suites (Greek translations of MMLU, ARC, HellaSwag, WinoGrande and Belebele [2]) are the same instruments our retention suite (§10) inherits. Sophea-Titan-1, the model whose instruction corpus is our direct half, belongs to that same line, as do our two companion efforts: a Greek adaptation of the Nemotron retrieval stack with HERA, a large-scale Greek retrieval-augmented-generation benchmark [18], and MORFES, an expert-verified benchmark for productive Greek inflectional morphology [23]. The threats to the axes we evaluate *on* are equally documented, and two of ours (§8) are new exemplars of known mechanisms: Artetxe et al. [1] show that translating premise and hypothesis independently reduces lexical overlap in NLI (our degrading category), and Singh et al. [32] show that rankings change depending on whether models are evaluated on the full or the culturally-sensitive subset of a translated benchmark. What §8 adds to that literature is a distinct mechanism: the artifact can favour the *fine-tuned* model rather than distorting all models alike, when the benchmark translator and the training-data generator are the same model family. Degenerate repetition under greedy decoding is the classic result of Holtzman et al. [15]; our contribution is negative, that it is not separable from length in this setting.

Limitations

Every model here is a sparse MoE, and every one is adapted with LoRA. Every result here is therefore a statement about *LoRA on mixture-of-experts models*, not about supervised fine-tuning in general. Two specific reasons this could matter rather than being boilerplate: the adapters touch only a strided subset of MoE layers plus the shared expert, so a dense model (where LoRA reaches every FFN) may behave differently; and the seed sensitivity we report (§4) has a plausible MoE-specific mechanism, since routing decides which experts receive gradient at all, and a different initialisation can send a different subset of experts down a different path. We did

not test a dense baseline, and cannot separate “LoRA SFT is seed-sensitive at this scale” from “sparse routing amplifies seed sensitivity”. A dense control is the single cheapest experiment that would sharpen this paper.

Single seed for all but one configuration; the noise floor is estimated from $n=3$ on one arm and $n=2$ on another. Four conditions were never varied: LoRA rank and targets, one epoch, the fixed non-reasoning arm in every two-phase mix, and greedy decoding. All three axes are Greek versions of English benchmarks; a natively-authored Greek commonsense probe we built came out at ceiling (96–97% for every arm) and could not discriminate. The logic axis is the smallest after decontamination and carries the seed instability, and every arm is near-blind to one of its three classes (14–16% recall on *False*), a failure a single accuracy number conceals.

Greek is mid-resource, and the auditability claim is a hypothesis. Greek is an EU official language with dedicated open models; whether this recipe transfers to truly low-resource languages (which lack the strong base and the language-directed corpus it depends on) is untested. And no human read the traces: apart from the ~ 150 -trace hand-labelled switching probe of §2 itself (one annotator, no guidelines or agreement measurement), every fidelity number in this paper is automatic, so “a trace the user can read and audit” is measured here only as script identity (the small LLM-judged register and grammar probes of §6 are the sole exception), not as fluency, terminology quality, or followability; and it inherits the standing assumption that a chain-of-thought trace is faithful to the computation it narrates. Two cheaper alternatives to SFT were not compared: translating the base’s English trace post hoc, and few-shot Greek-trace exemplars (our “prompting cannot reach it” evidence covers bare instructions only). Finally, the zero-switch target treats any code-switching as a defect; Greek technical registers routinely borrow English terms, and we did not measure what the monolingual constraint costs in terminology fidelity.

Availability

We release five fine-tuned checkpoints across the three families of Table 1: **Sophea-Qwen3.6-v1**, **Sophea-OSS-v1**, **Sophea-Nemo-3-Nano-v1** and **Sophea-Nemo-3.5-Lightning-v1** (the four SFT releases), plus **Sophea-Qwen3.6-v1.1**, the RLVR override arm of §15.1, which passed every hold-gate (fidelity 98.27, both held directions kept; the 60% override target itself was missed). The Qwen releases ship the base’s full multimodal (vision) stack and its multi-token-prediction head for speculative decoding; the Lightning release ships its base’s MTP head as well. All five ship in the **Sophea Reasoning Models** collection

at <https://huggingface.co/collections/KIEFERSA/sophea-reasoning-models>.

The controls are the methodology we would most like reused (§16 and Table 15 carry the full list). Each is a few hours of compute, none needs our models or our corpus, and each changed a conclusion we had already written down; run in the order given they cost less than one ablation and would have saved us most of a project.

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